

Where the character values of $\mathrm{Sp}(2m)$ at a torsion element fail to generate the ring of integers

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Abstract

Let $g \in \mathrm{Sp}(2m)$ have eigenvalues $\xi_q^{\pm 1}, \dots, \xi_q^{\pm m}$, with $1 \leq m < q/2$. The values of all characters of $\mathrm{Sp}(2m)$ at g generate a subring A of the ring of integers $\mathcal{O}$ of $\mathbb{Q}(\xi_q)^+$, which, apart from known cases where it is $\mathbb{Z}$, is an order. We locate and describe its conductor. Write $h = 2m$ and $q = p^v q'$ with $p \nmid q'$. Every prime of the index $[\mathcal{O} : A]$ divides q , and for $q' \notin \{1, 2, 3, 4, 6\}$ a prime $p \mid q$ divides it if and only if q' divides h , $h + 1$ or $h + 2$. At every prime the conductor is a power of the radical. At such a p with $\varphi(p^v) \geq 2$, the first graded layer of A has the dimension of the cyclic module over $\mathbb{F}_p[(\mathbb{Z}/q')^\times]$ generated by an explicit first-moment function S , and it decides whether that power is 1 and, when it is 2, both lengths and whether the local ring is Gorenstein. For p odd and $q' \notin \{1, 2, 3, 4, 6\}$, the stable segments give S two shapes: Stickelberger's sawtooth when $m \equiv 0, -1 \pmod{q'}$, and that sawtooth shifted by half a period when $2m + 1 = Nq'$ or when q' is even and $m \equiv q'/2, q'/2 - 1 \pmod{q'}$. On the sawtooth families and on the half-period ones with q' even, a half-period identity and a doubling lemma put the two on one lattice, and Sinnott's index formula computes the p -part of the maximal minors of the moment matrix: the conductor at p is the radical exactly when p does not divide $h^-(\mathbb{Q}(\zeta_{q'}))$, provided $p \nmid \lceil m/q' \rceil$ (on the half-period families, $p \nmid \lceil 2m/q' \rceil$), and a simple divisor of h^- gives the square of the radical at p and a Gorenstein local ring. When instead $2m + 1 = Nq'$ and $p \nmid N h^-$, the first layer, both lengths and the Cohen–Macaulay type are read off the subgroup generated by 2 in $(\mathbb{Z}/q')^\times$. Whenever that first layer, of dimension w , is more than half full, the index at p is settled: $v_p([\mathcal{O} : A]) = 2d - 1 - w$, with $d = \varphi(q')/2$ the dimension of the residue ring. The answers bring together Dedekind's conductor, Kummer's link between Bernoulli numbers and class numbers, and the combinatorics of musical scales.

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1 What this note is about

A character of a compact Lie group, evaluated at an element of finite order, is an algebraic integer. Take all characters at once, at one element, and they generate a ring. For the element $g = g_{1/q} \in \mathrm{Sp}(2m)$ with eigenvalues $\xi_q^{\pm j}$, $1 \leq j \leq m$, that ring sits inside the ring of integers $\mathcal{O} = \mathbb{Z}[\xi_q + \xi_q^{-1}]$ of the real cyclotomic field. Two questions follow, one inside the other:

Do the character values generate the ring of integers? And where they do not, what exactly is missing?

Both questions are older than the ring. In 1877, counting ideal classes in orders of a number field, Dedekind attached to each order the largest ideal of the maximal order that it contains, its *Führer*, or conductor [12]. A year later he showed that above a prime p not dividing the

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index $[\mathcal{O} : \mathbb{Z}[\alpha]]$ the primes of $\mathcal{O}$ can be read off the factorisation of the minimal polynomial of α modulo p [13]. Which primes divide the index, and what the conductor looks like at each of them, have been the first things to ask about an order ever since. What changes here is who chooses the generators: not one number α , but every irreducible character of $\mathrm{Sp}(2m)$, evaluated at once at g .

The first question has a clean answer, and the answer to the second turns out to be written in three older languages: the combinatorics of musical scales, the length formalism of one-dimensional rings, and the arithmetic of Bernoulli numbers. The note is organised as a sequence of questions, each one raised by the answer to the previous one. Figure 1 draws that thread, and the table lists the answers.

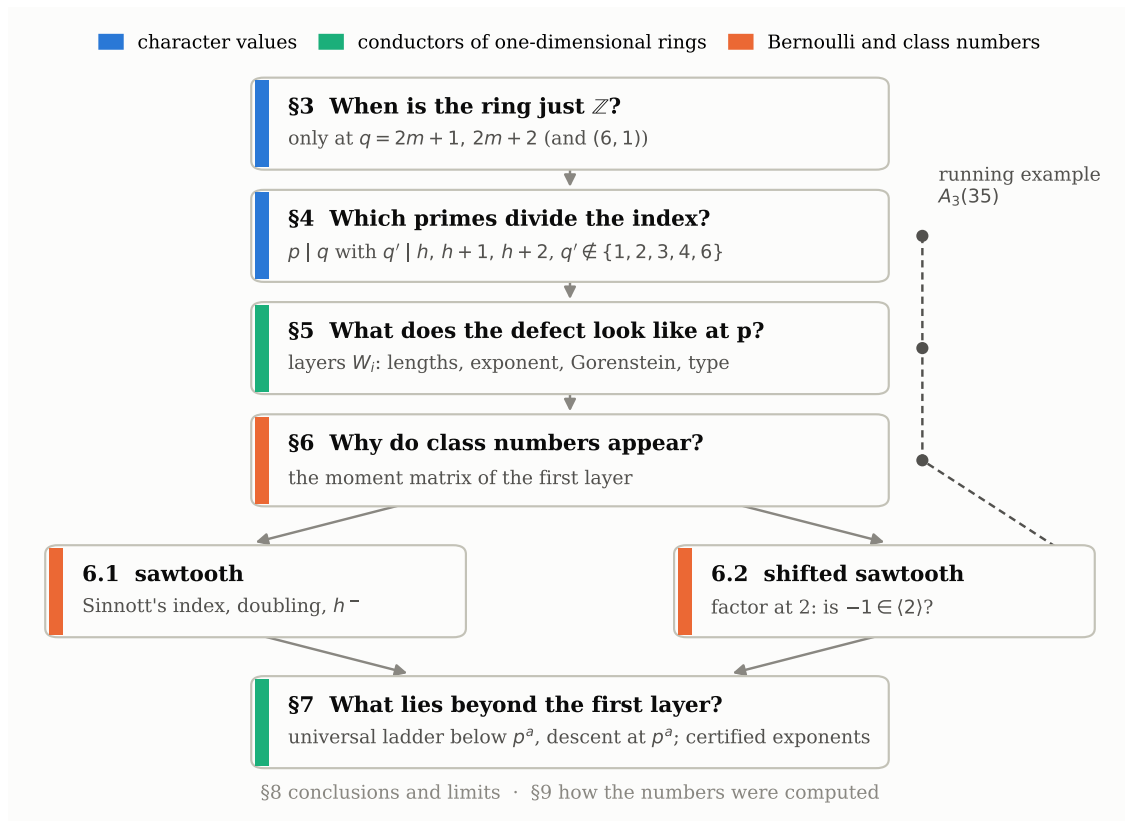


Figure 1: The thread of the note. Each section answers the question left by the one above it; at §6 the two kinds of stable segment split the answer in two. The colour of each box marks which of the three lines of §2 it draws on, and the dashed line follows the running example $A_3(35)$.

question	answer
When is the ring just $\mathbb{Z}$? (§3)	exactly at the integer points of an earlier note: Proposition 2
Which primes divide the index? (§4)	only primes of q ; and, for $q' \notin \{1, 2, 3, 4, 6\}$, $p \mid [\mathcal{O} : A]$ iff the prime-to- p part q' of the order of g divides h , $h + 1$ or $h + 2$: Theorem 4, Figure 3
What does the defect look like at one prime? (§5)	a single residue field $\mathbb{F}_p$, a first layer measured by a cyclic module, lengths as sums of layer defects, a Gorenstein criterion and the Cohen–Macaulay type: Theorem 5, Propositions 6–11
Why do class numbers appear? (§6)	the maximal minors of the moment matrix are Sinnott’s index times $(2q')^d/w_{q'}$, a unit at odd $p \nmid q'$ (read at $q'/2$ when $q' \equiv 2 \pmod{4}$), so h^- decides whether the first layer is full; for $2m + 1 = Nq'$ and $p \nmid N h^-$ the subgroup $\langle 2 \rangle$ decides it: Proposition 13, Lemma 14, Theorem 12, Corollaries 16, 17, 18, Figures 5–8
What lies beyond the first layer? (§7)	a universal ladder below $p^{v_p(N)}$, descent to a smaller order at $p^{v_p(N)}$, exponents certified by semigroups, and a question on what lies beyond: Propositions 19, 20, Corollary 7, Figure 9
What is proved, and where does it stop? (§8)	what is ours, what rests on others, what was only measured, and what is not covered

A running example. Take $m = 3$ and $q = 35$, so $A = A_3(35)$ has rank $\varphi(35)/2 = 12$ and $h = 6$. At $p = 7$, $q' = 5$ divides none of 6, 7, 8; at $p = 5$, $q' = 7 = h + 1$. So only 5 divides the index, and the example is followed through §§4–6 until $[\mathcal{O} : A] = 5^3$ and $\mathfrak{f}\mathcal{O}_5 = \mathfrak{r}^2$ come out by hand.

symbol	meaning
$q, \xi, K, \mathcal{O}$	order of g ; $\xi = \xi_q$; $K = \mathbb{Q}(\xi)^+$; $\mathcal{O} = \mathbb{Z}[\xi + \xi^{-1}]$
m, h	rank; $h = 2m$, the Coxeter number of C_m
$A, \mathfrak{f}$	the order (1); its conductor $(A : \mathcal{O})$
p, v, q', E	a prime; $q = p^v q'$, $p \nmid q'$; $E = \varphi(p^v)$, the ramification index of p in K when $q' \geq 3$
D	$[K : \mathbb{Q}] = \varphi(q)/2$; $D = Ed$ when $q' \geq 3$
f, d	residue degree of p in K ; $d = \varphi(q')/2 = \dim_{\mathbb{F}_p} \mathcal{O}/\mathfrak{r}$
$\mathfrak{r}, e = e_p$	radical of $\mathcal{O}_p = \mathcal{O} \otimes \mathbb{Z}_p$; $\mathfrak{f}\mathcal{O}_p = \mathfrak{r}^e$
$N^{(n)}, S$	the segment $\{\pm 1, \dots, \pm m\}$ counted mod n ; its first moment (2)
W_i	$\dim_{\mathbb{F}_p}$ of the i -th graded layer of A at p
N	the quotient in $m = Nq'$, $m + 1 = Nq'$ or $2m + 1 = Nq'$

Theorems are boxed and key formulas framed. Statements that rest on computation rather than proof are labelled *measured* or *observed*, with the count and the range. In one place, the answers read as follows.

Main results. Let $A \neq \mathbb{Z}$, let p be a prime, $q = p^v q'$ with $p \nmid q'$, $E = \varphi(p^v)$ and $d = \varphi(q')/2$.

- (1) *Support* (Theorem 4). If $p \mid [\mathcal{O} : A]$ then $p \mid q$; and if $p \mid q$ and $q' \notin \{1, 2, 3, 4, 6\}$, then $p \mid [\mathcal{O} : A]$ if and only if q' divides h , $h + 1$ or $h + 2$.
- (2) *Local structure* (Lemma 3, Theorem 5). $\mathfrak{f} \mathcal{O}_p = \mathfrak{r}^{e_p}$. If moreover $q' \notin \{1, 2, 3, 4, 6\}$, $E \geq 2$ and $p \mid [\mathcal{O} : A]$, then $W_1 = \dim_{\mathbb{F}_p} \mathbb{F}_p[(\mathbb{Z}/q')^\times] \cdot S$ with S as in (2); $e_p = 1$ if and only if $W_1 = d$; $v_p([\mathcal{O} : A]) = \sum_{i < e_p} (d - W_i)$, which for $W_1 > d/2$ is $v_p([\mathcal{O} : A]) = 2d - 1 - W_1$ (Corollary 18), and $v_p([A : \mathfrak{f}]) = \sum_{i < e_p} W_i$; and A_p is Gorenstein if and only if $2 \sum_{i < e_p} W_i = e_p d$ (Proposition 10).
- (3) *Class numbers* (Theorem 12, Corollary 16). If p is odd, $p \mid q$, $q' \notin \{1, 2, 3, 4, 6\}$, $m \equiv 0, -1 \pmod{q'}$ and $p \nmid [m/q']$, then $d - W_1$ is the number of elementary divisors of the moment matrix $M_{q'}$ divisible by p ; so $e_p = 1$ if and only if $p \nmid h^-(\mathbb{Q}(\zeta_{q'}))$, and if $v_p(h^-) = 1$ then $e_p = 2$ and A_p is Gorenstein. For even $q' \geq 8$ the same holds on the half-period families $m \equiv r, r - 1 \pmod{q'}$, $r = q'/2$, with $p \nmid [m/r]$ (Lemma 14). If $q' \geq 5$ is prime and $2m + 1 = Nq'$ with $p \nmid N(q' - 1)$, then $d - W_1$ counts the odd characters χ with $(\bar{\chi}(2) - 1)B_{1,\chi} \equiv 0 \pmod{\mathfrak{p}}$.
- (4) *The factor at 2* (Corollary 17). If p is odd, $p \mid q$, $q' \geq 5$ is odd, $2m + 1 = Nq'$ and $p \nmid N h^-(\mathbb{Q}(\zeta_{q'}))$, then $e_p = 1$ if $-1 \in \langle 2 \rangle \subseteq (\mathbb{Z}/q')^\times$; otherwise, with t the order of 2 and $\delta = d/t$, $W_1 = d - \delta$, $e_p = 2$, $v_p([\mathcal{O} : A]) = d + \delta - 1$, and A_p is almost Gorenstein of type $2\delta - 1$.

2 Three lines that meet

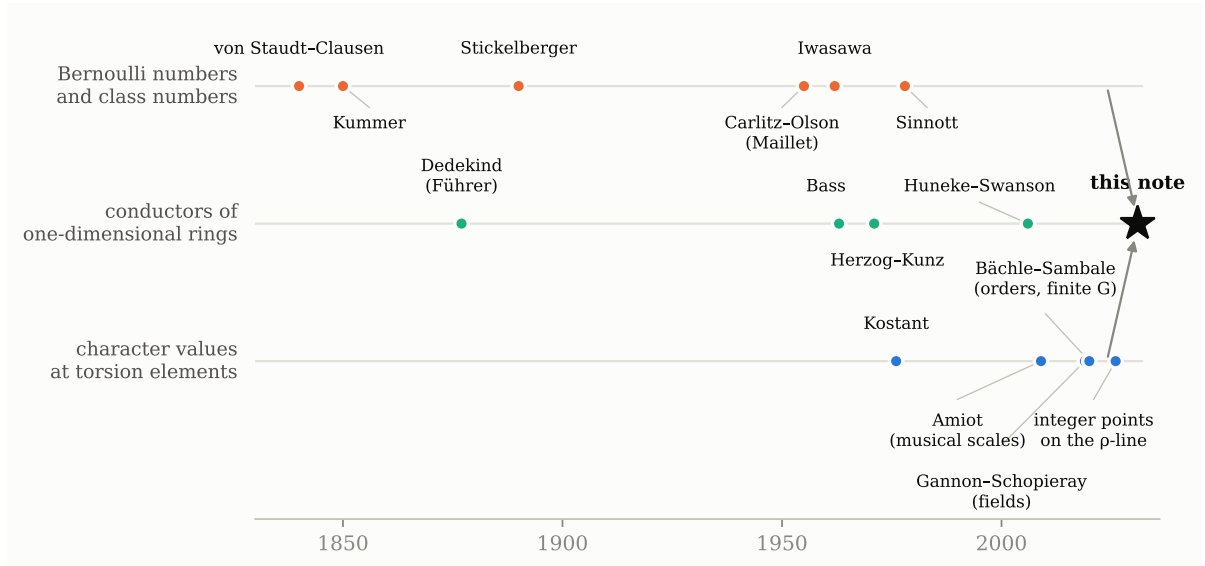


Figure 2: Three lines of work, and the point where this note joins them. Dates are years of publication (for Gannon–Schopieray, of the preprint).

Bernoulli numbers and class numbers. The numbers B_k enter number theory with

von Staudt and Clausen, who located their denominators, and with Kummer, who tied the divisibility of their numerators by a prime p to the class number of $\mathbb{Q}(\zeta_p)$. Stickelberger's element annihilates the class group; Iwasawa and Sinnott computed the index of the ideal it generates, which is the relative class number h^- up to a power of 2 [25, 29]. In between, Maillet asked for the determinant of the matrix of least residues $R(rs^{-1})$, and Carlitz and Olson showed that it is a power of p times h^- [10].

Conductors of one-dimensional rings. For a reduced one-dimensional ring A with finite normalisation $\mathcal{O}$, the conductor $\mathfrak{f} = (A : \mathcal{O})$, Dedekind's *Führer*, measures how far A is from being normal. Bass recorded, after Apéry, Gorenstein and Samuel, that a Gorenstein ring satisfies $\ell(\mathcal{O}/A) = \ell(A/\mathfrak{f})$ [7, Cor. (6.5)]; Herzog and Kunz, and later Huneke and Swanson, put the corresponding inequality and its equality case into the theory of the canonical module [16, 17].

Character values at torsion elements. Kostant singled out elements at which every character of a simple group takes one of the values $0, \pm 1$ [18, Thm. 3.1]; for $\mathrm{Sp}(2m)$ one of them, the principal element of type ρ , is $g_{1/(2m+2)}$. The fields generated by character values are understood in the modular setting [15], and for finite groups Bächle and Sambale studied the *orders* $\mathbb{Z}[\chi(g) : \chi \in \mathrm{Irr}(G)]$ [6]. For $\mathrm{Sp}(2m)$ along the line $g_{1/q}$, the points at which every character is an integer were classified in [24]; a map of the older history of that question is drawn there. Finally, from music theory: which intervals generate a given scale in $\mathbb{Z}/c$ was settled by Amiot [1], and that is exactly the combinatorial lemma this note needs.

3 When is the ring just $\mathbb{Z}$?

Before asking what is missing, one should know when nothing is there to miss.

Fix $q \geq 3$, $1 \leq m < q/2$, $\xi = \xi_q$, $\eta_j = \xi^j + \xi^{-j}$, and $\mathcal{O} = \mathbb{Z}[\xi + \xi^{-1}]$, the ring of integers of $K = \mathbb{Q}(\xi)^+$. The representation ring of $\mathrm{Sp}(2m)$ is $\mathbb{Z}[e_1, \dots, e_m]$, with e_i the elementary symmetric functions of $x_1 + x_1^{-1}, \dots, x_m + x_m^{-1}$; hence

$$A = A_m(q) = \mathbb{Z}[e_1(\eta_1, \dots, \eta_m), \dots, e_m(\eta_1, \dots, \eta_m)] \subseteq \mathcal{O} \quad (1)$$

Everything depends on how the segment $\{\pm 1, \dots, \pm m\}$ sits modulo an integer n . Let $N^{(n)}(c) = \#\{1 \leq j \leq m : j \equiv \pm c \pmod{n}\}$ on classes $c \in (\mathbb{Z}/n)/\pm 1$.

Lemma 1 (segments). *Let $n \geq 3$ and let a be a unit mod n with $a \not\equiv \pm 1$. Then $N^{(n)}$ is stable under $c \mapsto ac$ if and only if n divides h , $h+1$ or $h+2$; in that case it is stable under every unit.*

Proof. Write $m = kn + r$ with $0 \leq r < n$. Complete periods contribute a unit-invariant multiset, so stability is that of the segment $\{\pm 1, \dots, \pm s\}$ with $s = r$ if $r \leq n/2$ and $s = n-1-r$ otherwise. The set $[-s, s] \subseteq \mathbb{Z}/n$ is a scale generated by 1; if $a[-s, s] = [-s, s]$ it is also generated by a , and by [1], Theorem 1, a coprime generator other than ± 1 forces the scale to have 1, $n-1$ or n elements. These are $r \in \{0, n-1\}$, $r = (n-1)/2$ for odd n , and $r \in \{n/2-1, n/2\}$ for even n ; that is, $n \mid h$, $n \mid h+2$ or $n \mid h+1$. Those segments are stable under every unit. The same classification, with another proof, is in [24], Theorem 5.1. $\square$

The musical reading is literal. In $\mathbb{Z}/12$ the diatonic scale $\{0, 2, 4, 5, 7, 9, 11\}$ is a run of seven consecutive fifths, the scale Pythagorean tuning stacks: a segment for the generator 7. It has seven notes, neither 1, 11 nor 12, so by Amiot's theorem no generator other than ± 7 lays it out as a segment. In the lemma the roles are swapped: the segment $\{\pm 1, \dots, \pm m\}$ is laid out by 1, and the question is which other generators it admits. The answer is governed by $h = 2m$, the Coxeter number of type C_m , that is, the order of the product of the simple reflections of its Weyl group.

family	$m \bmod n$	the segment $\{\pm 1, \dots, \pm m\} \bmod n$	example ($m = 6$)
$n \mid h$	0 or $n/2$	whole periods (+ the class $n/2$)	$n = 3, 4, 6, 12$
$n \mid h + 1$	$(n - 1)/2$	every nonzero class, once more	$n = 13$
$n \mid h + 2$	$n - 1$ or $n/2 - 1$	whole periods minus $\{0\}$ (and $\{n/2\}$)	$n = 7, 14$

Table 1: The three stable families. $h = 2m$ is the Coxeter number of type C_m .

Proposition 2 ($A = \mathbb{Z}$). $A = \mathbb{Z}$ if and only if $q \in \{2m + 1, 2m + 2\}$ or $(q, m) = (6, 1)$. Otherwise A is an order of rank $\varphi(q)/2$ in K .

Proof. The fraction field of A is the fixed field of the stabiliser of $N^{(q)}$ in $\text{Gal}(K/\mathbb{Q}) = (\mathbb{Z}/q)^\times / \pm 1$. For $q \notin \{3, 4, 6\}$ Lemma 1 makes that stabiliser trivial or everything, and with $2m < q$ it is everything exactly for $q \in \{2m + 1, 2m + 2\}$. For $q \in \{3, 4, 6\}$, $\mathcal{O} = \mathbb{Z}$; with $2m < q$ this gives $m = 1$ for $q = 3, 4$ and $m \in \{1, 2\}$ for $q = 6$. $\square$

This is not new. For $m \geq 2$ it is the part $2m < q$ of [24], Theorem 5.1, whose statement also contains a sporadic family $q = 6$, $m \equiv 1 \pmod{3}$, outside the range $2m < q$ here; the case $(6, 1)$ is its $m = 1$ shadow. In the language of rational elements it says that $g_{1/q}$ is conjugate to all its prime-to- q powers exactly at these points, and for $q = 2m + 2$ the element is one of Kostant's [18, Thm. 3.1]: his element $\exp(2\pi i \cdot 2\rho)$, with the invariant form normalised by $(\theta, \theta) = 1/h^\vee$, is $g_{1/(2m+2)}$ for $\text{Sp}(2m)$ (§9). From now on $A \neq \mathbb{Z}$ and $\mathfrak{f} = (A : \mathcal{O})$.

Lemma 3 (Galois stability). Assume $A \neq \mathbb{Z}$. The order A and its conductor $\mathfrak{f}$ are stable under $\text{Gal}(K/\mathbb{Q})$. Hence, for every prime p , all primes of $\mathcal{O}$ above p occur in $\mathfrak{f}$ with the same exponent $e_p \geq 0$: $\mathfrak{f}\mathcal{O}_p = \mathfrak{r}_p^{e_p}$, where $\mathfrak{r}_p$ is the radical of $\mathcal{O}_p = \mathcal{O} \otimes \mathbb{Z}_p$.

Proof. Let $C_u \in \mathbb{Z}[X]$ be Chebyshev's polynomial, rescaled so that $C_u(2 \cos \theta) = 2 \cos u\theta$; equivalently $C_u(t + t^{-1}) = t^u + t^{-u}$, with $C_0 = 2$, $C_1 = X$, $C_{u+1} = XC_u - C_{u-1}$. For $\sigma_u(\xi) = \xi^u$ one has $\sigma_u(\eta_j) = C_u(\eta_j)$, so $\sigma_u(e_k(\eta_1, \dots, \eta_m)) = e_k(C_u(\eta_1), \dots, C_u(\eta_m))$, a symmetric polynomial with integer coefficients in $\eta_1, \dots, \eta_m$, hence an element of A . Thus $\sigma_u(A) \subseteq A$ for every u , and equality follows by applying σ_u^{-1} . Then $\sigma_u(\mathfrak{f}) = (\sigma_u(A) : \sigma_u(\mathcal{O})) = \mathfrak{f}$, and the primes above p form one Galois orbit. $\square$

▷ The integer points $q \mid h, h + 1, h + 2$ belong to type C . For another simply connected simple group, at which orders q is $\exp(2\pi i \rho/q)$ conjugate, with ρ carried into the Cartan subalgebra by the invariant form normalised as in type C (where this element is $g_{1/q}$), to all its prime-to- q powers, and can those q be read off the regular numbers of its Weyl group?

4 Which primes divide the index?

The ring A has finite index in $\mathcal{O}$. Is there any pattern in the primes of that index?

For a prime p write $q = p^v q'$ with $p \nmid q'$, $E = \varphi(p^v)$, and let f be the order of p in $(\mathbb{Z}/q')^\times / \pm 1$, the residue degree of p in K . Define, for $c \in \mathbb{Z}/q'$,

$$S(c) = \sum_{\substack{1 \leq j \leq m \\ j \equiv c \pmod{q'}}} j - \sum_{\substack{1 \leq j \leq m \\ j \equiv -c \pmod{q'}}} j \quad (2)$$

an odd function on $\mathbb{Z}/q'$: the first moment of the segment, class by class.

Theorem 4 (support of the conductor). Assume $A \neq \mathbb{Z}$.

- (a) If $p \nmid q$, then $p \nmid [\mathcal{O} : A]$.
- (b) Let $p \mid q$ and $q' \geq 3$. Then $p \mid [\mathcal{O} : A]$ if and only if at least one of the following fails: (i) the stabiliser of $N^{(q')}$ in $(\mathbb{Z}/q')^\times / \pm 1$ is trivial; (ii) $E = 1$, or $S(c) \not\equiv 0 \pmod{p}$ for some c with $2c \not\equiv 0 \pmod{q'}$.
- (c) If $p \mid q$ and $q' \notin \{1, 2, 3, 4, 6\}$, then

$$p \mid [\mathcal{O} : A] \iff q' \mid h, \quad q' \mid h+1 \quad \text{or} \quad q' \mid h+2.$$

Read through a prime $\mathfrak{P}$ of $\mathbb{Z}[\xi]$ above p , part (c) says: reduction kills the p -power part of g and leaves its prime-to- p part, of order q' ; and for $q' \notin \{1, 2, 3, 4, 6\}$, p divides the index exactly when that part lies on one of the three families $q' \mid h, h+1, h+2$ of integer points ([24], Theorem 5.1; Proposition 2 is their part with $2m < q$). For the running example $A_3(35)$: at $p = 7$ the part $q' = 5$ divides none of $h, h+1, h+2 = 6, 7, 8$, so $7 \nmid [\mathcal{O} : A]$; at $p = 5$, $q' = 7 = h+1$, so $5 \mid [\mathcal{O} : A]$. Figure 3 shows the index over the (q, m) -plane, split by prime and coloured by family.

Proof. Since $[\mathcal{O} : A]$ is finite, $p \nmid [\mathcal{O} : A]$ if and only if $A + p\mathcal{O} = \mathcal{O}$, i.e. the image of A in $\mathcal{O}/p\mathcal{O}$ is everything.

(a) For $p \nmid q$, $\mathcal{O}/p\mathcal{O}$ is étale and its $\overline{\mathbb{F}}_p$ -points are the reductions of $\xi \mapsto \xi^u$, $u \in (\mathbb{Z}/q)^\times / \pm 1$. At the point u the e_i are the elementary symmetric functions of $u \cdot N^{(q)}$, and distinct multisets give distinct points because $c \mapsto \omega^c + \omega^{-c}$ is injective on classes for a primitive root ω of order q in characteristic p . So A generates $\mathcal{O}/p\mathcal{O}$ iff the stabiliser of $N^{(q)}$ is trivial, iff $A \neq \mathbb{Z}$ (Lemma 1, Proposition 2). For the orders generated by all irreducible characters of a finite group G , the analogous fact, that the primes of the index divide $|G|$, is [6], Proposition 6; it does not apply to A directly, since A is generated by characters of $\mathrm{Sp}(2m)$ only.

(b) In $\mathbb{Z}_p[\xi]$ write $\xi = \omega(1 + \pi)$ with ω of order q' and $1 + \pi$ of order p^v ; then $\mathbb{Z}[\xi]/p = \mathbb{F}_p[\omega][\pi]/(\pi^E)$. As $q' \geq 3$, complex conjugation is not in the inertia group, so $\mathcal{O}/p\mathcal{O} \cong \prod_{P|p} \mathbb{F}_{p^f}[z]/(z^E)$ with $z = \eta_1 - (\omega + \omega^{-1}) \equiv (\omega - \omega^{-1})\pi$. Let $\mathfrak{r}$ be the radical and $\bar{R} = \mathcal{O}/\mathfrak{r}$ the residue ring. If $E = 1$, then $\mathcal{O}/p\mathcal{O} = \bar{R}$ and only (i) matters: the argument of (a) applies with q' in place of q . Assume $E \geq 2$, and let B_2 be the image of A in $\Lambda_2 = \mathcal{O}/(p, \mathfrak{r}^2) = \bar{R}[z]/(z^2)$. If $B_2 = \Lambda_2$, then the image of A in $\mathcal{O}/p\mathcal{O}$ contains $\bar{R}$ (through $x \mapsto x^{p^{fk}}$ with $p^{fk} \geq E$, which fixes $\bar{R}$ and kills z) and a generator of $\mathfrak{r}/\mathfrak{r}^2$, hence everything; the converse is clear. Modulo $\mathfrak{r}^2$, $\eta_j \equiv \gamma_j + jU_j z$ with $\gamma_j = \omega^j + \omega^{-j}$ and $U_j = (\omega^j - \omega^{-j})/(\omega - \omega^{-1})$, so $\prod_j (T - \eta_j) \equiv F(T) - zG(T)$ with $F = \prod_j (T - \gamma_j)$ and

$$G = F \sum_j \frac{jU_j}{T - \gamma_j} = F \sum_c \frac{S(c)U_c}{T - \gamma_c}, \quad (3)$$

the second sum over classes with $N^{(q')}(c) \geq 1$. The coefficients of F generate $\bar{R}$ iff the stabiliser of $N^{(q')}$ is trivial, as in (a). In that case $\bar{R} \subseteq B_2$, and the z -part of B_2 is the $\bar{R}$ -submodule generated by the coefficients of G , which is all of $z\bar{R}$ iff G is nonzero in every residue field. The functions $U_c F/(T - \gamma_c)$ with $2c \not\equiv 0$ and $N^{(q')}(c) \geq 1$ are linearly independent, so this happens iff $S(c) \not\equiv 0 \pmod{p}$ for some such c .

(c) By Lemma 1, (i) fails iff $q' \mid h, h+1, h+2$. When (i) holds, $m = kq' + r$ with r outside

the list of that lemma, and counting gives, for $1 \leq c \leq q' - 1$,

$$S(c) = \begin{cases} (2k+1)c & c \leq r < q' - c, \\ -(2k+1)(q' - c) & q' - c \leq r < c, \\ (k+1)(2c - q') & q' - c \leq r \text{ and } c \leq r, \\ k(2c - q') & r < c \text{ and } r < q' - c. \end{cases}$$

Write $n = q'$. Since $r \notin \{0, n-1\}$, the residue $c = 1$ lies in the first case and $S(1) = 2k+1$. If $r < n/2$ (so $r < (n-1)/2$ for odd n and $r < n/2 - 1$ for even n), take $c = (n-1)/2$ for odd n , which gives $S(c) = -k$, or $c = n/2 - 1$ for even n , which gives $S(c) = -2k$. If $r > n/2$, take $c = (n+1)/2$ or $c = n/2 + 1$, which gives $S(c) = k+1$ or $2(k+1)$. In every case p cannot divide both $S(1) = 2k+1$ and $S(c)$, since $\gcd(2k+1, k) = \gcd(2k+1, k+1) = 1$. So (ii) holds whenever (i) does, for $p = 2$ as well. $\square$

▷ *Theorem 4(b) has two clauses: an étale one, on the Galois stabiliser of the prime-to- p part of g , and a tangent one, on the first moment. For a torsion element of any simply connected group, is $p \mid [\mathcal{O} : \mathbb{Z}[\chi(g)]]$ still the failure of one of the two, with the differential of the fundamental characters in the role of S ?*

5 What does the defect look like at oneprime?

Knowing that p divides the index, one would like to see the conductor at p : how many maximal ideals, which residue fields, how deep.

Assume $p \mid q$, $q' \notin \{1, 2, 3, 4, 6\}$, $E \geq 2$, and $q' \mid h, h+1, h+2$, so that $p \mid [\mathcal{O} : A]$ and $N^{(q')}$ is stable under every unit. Let $\mathfrak{r}$ be the radical of $\mathcal{O}_p = \mathcal{O} \otimes \mathbb{Z}_p$, $A_p = A \otimes \mathbb{Z}_p$, $d = \varphi(q')/2 = \dim_{\mathbb{F}_p} \bar{R}$, and let $\text{gr}_i(A)$ be the image of $A_p \cap \mathfrak{r}^i$ in $\mathfrak{r}^i/\mathfrak{r}^{i+1} \cong \bar{R}$. Put $W_i = \dim_{\mathbb{F}_p} \text{gr}_i(A)$, so that $W_0 = 1$ and $W_i \leq d$; by Lemma 3, $\mathfrak{f}\mathcal{O}_p = \mathfrak{r}^e$ for a single exponent $e = e_p$.

Theorem 5 (local structure). *Under these hypotheses:*

- (a) *the image of A in $\bar{R}$ is $\mathbb{F}_p$; so A has exactly one maximal ideal above p , it contains $\mathfrak{f}$, and its residue field is $\mathbb{F}_p$;*
- (b) *$W_1 = \dim_{\mathbb{F}_p} \mathbb{F}_p[(\mathbb{Z}/q')^\times] \cdot S$, where $(u \cdot S)(c) = S(u^{-1}c)$;*
- (c) *$e = 1$ if and only if $W_1 = d$;*
- (d) *$v_p([\mathcal{O} : A]) = \sum_{i=0}^{e-1} (d - W_i)$ and $v_p([A : \mathfrak{f}]) = \sum_{i=0}^{e-1} W_i$; in particular, if $e = 2$, then $v_p([A : \mathfrak{f}]) = 1 + W_1$ and $v_p([\mathcal{O} : A]) = 2d - 1 - W_1$.*

Proof. (a) Stability makes $F \in \mathbb{F}_p[T]$: the e_i reduce to the same constants of $\mathbb{F}_p$ in every residue field. By Theorem 4, $p \mid [\mathcal{O} : A]$, so that ideal contains $\mathfrak{f}$.

(b) Every element of A is $\Phi(e)$ with $\Phi \in \mathbb{Z}[X]$; modulo $\mathfrak{r}^2$, $\Phi(e) \equiv \Phi(F) + z \sum_i (\partial_i \Phi)(F) G_i$ with $\Phi(F) \in \mathbb{F}_p$, and $p \in \mathfrak{r}^2$. Hence $\text{gr}_1(A)$ is the $\mathbb{F}_p$ -span V of the coefficients G_i . Since $\bar{R}$ is étale, $\bar{R} \otimes_{\mathbb{F}_p} \bar{\mathbb{F}}_p \cong \bar{\mathbb{F}}_p^d$, the coordinates being the d embeddings $\sigma_u: \omega + \omega^{-1} \mapsto \omega_0^u + \omega_0^{-u}$, $u \in (\mathbb{Z}/q')^\times / \pm 1$, for a fixed $\omega_0 \in \bar{\mathbb{F}}_p$ of order q' . Extension of scalars is injective on $\mathbb{F}_p$ -subspaces, so $\dim_{\mathbb{F}_p} V$ is the rank over $\bar{\mathbb{F}}_p$ of the matrix whose row u lists the coefficients of $\sigma_u(G)$. Multiplying row u by the unit U_u does not change that rank. From $\sigma_u(\gamma_c) = \gamma_{uc}$,

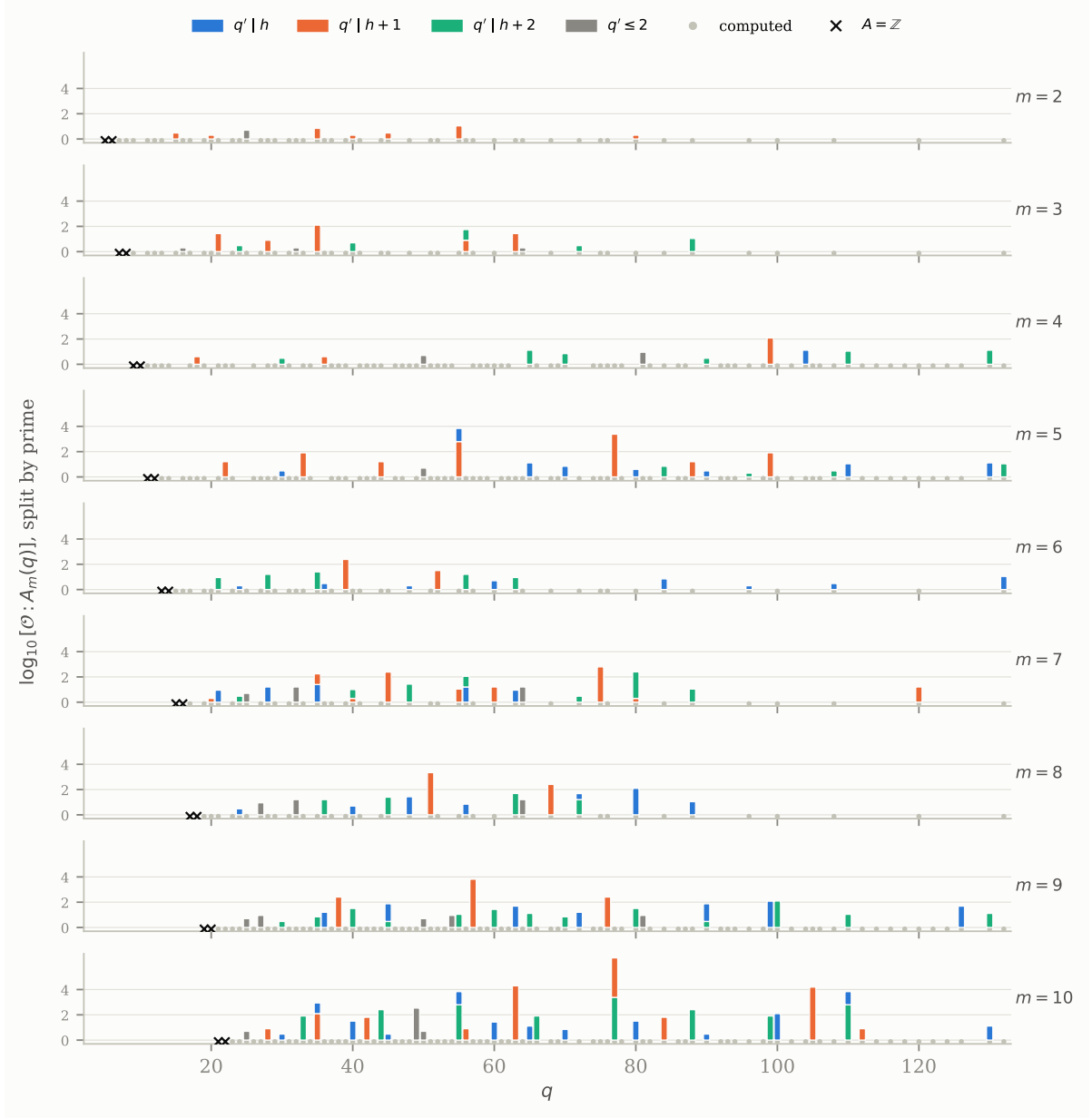


Figure 3: The index $[\mathcal{O} : A_m(q)]$ for $2 \leq m \leq 10$ and $q \leq 132$ (558 orders with a lattice computation, 156 of them with index > 1), one panel per m . Each bar is stacked by the primes p of the index, each segment of height $v_p \cdot \log_{10} p$ and coloured by the family of $q' = q/p^v$; grey segments are primes with $q' \leq 2$, outside the theorems. Dots mark computed q (no bar: index 1); crosses mark $A = \mathbb{Z}$. Every coloured segment lies over a pair with $q' \mid h, h+1, h+2$.

$\sigma_u(U_c) = U_{uc}/U_u$ and the stability of F ,

$$U_u \sigma_u(G) = \sum_c S(u^{-1}c) U_c \frac{F(T)}{T - \gamma_c}, \quad (4)$$

and the functions $U_c F/(T - \gamma_c)$, over the classes c with $2c \not\equiv 0 \pmod{q'}$ (for the others $U_c = 0$), are linearly independent over $\bar{\mathbb{F}}_p$ and do not depend on u . So the rank equals that of $[S(u^{-1}c)]_{u,c}$, a matrix with entries in $\mathbb{F}_p$, whose rank over $\bar{\mathbb{F}}_p$ equals its rank over $\mathbb{F}_p$, which is $\dim_{\mathbb{F}_p} \mathbb{F}_p[(\mathbb{Z}/q')^\times] \cdot S$.

(c) If $\text{gr}_1(A) = \text{gr}_1(\mathcal{O})$, then $\text{gr}_i(A) \supseteq \text{gr}_1(A)^i = \text{gr}_i(\mathcal{O})$ for all $i \geq 1$, since $\text{gr}(\mathcal{O}_p)$ is a polynomial ring over $\bar{R}$; so the closed sublattice $A_p \cap \mathfrak{r}$ has the graded of $\mathfrak{r}$ and equals it. The converse is clear.

(d) Filter $\mathcal{O}_p$ and A_p by $\mathfrak{r}^i$. For $i \geq e$, $A_p \cap \mathfrak{r}^i = \mathfrak{r}^i$, so only the layers $i < e$ contribute: $\mathcal{O}_p/A_p$ has a filtration with quotients of dimension $d - W_i$, and $A_p/\mathfrak{f}\mathcal{O}_p = A_p/\mathfrak{r}^e$ one with quotients of dimension W_i . $\square$

The running example. For $A_3(35)$ at $p = 5$: $E = 4$, $q' = 7$, $d = 3$, and since $m = 3 < 7$ the first moment is simply $S(c) = c$ for $c = 1, 2, 3$ (and $S(-c) = -S(c)$). With rows $u = 1, 2, 3$ and columns $c = 1, 2, 3$, and $u^{-1} \equiv 1, 4, 5 \pmod{7}$,

$$[S(u^{-1}c)] = \begin{pmatrix} 1 & 2 & 3 \\ -3 & 1 & -2 \\ -2 & 3 & 1 \end{pmatrix},$$

whose determinant is 0 and whose rank is 2 over $\mathbb{Q}$ and modulo 5. So $W_1 = 2 < d = 3$ and, by Theorem 5(c), the conductor at 5 is not the radical. That the matrix is singular over $\mathbb{Z}$ itself is explained in §6.

Products of layers make them usable in practice. Since $\text{gr}(A)$ is a subring of $\text{gr}(\mathcal{O}_p)$, the layers multiply: $\text{gr}_i(A) \cdot \text{gr}_j(A) \subseteq \text{gr}_{i+j}(A)$. Write $\bar{R} = \prod_{a=1}^g k_a$ and identify every layer with $\bar{R}$ through uniformisers z_a of the factors; by Lemma 3 the group $\text{Gal}(K/\mathbb{Q})$ permutes the factors transitively and preserves $A_p \cap \mathfrak{r}^i$, acting on each layer by a permutation of the factors, semilinear on each factor and twisted by units.

Proposition 6 (growth of layers). *Under the hypotheses of Theorem 5, let $s \geq 1$ with $W_s > 0$. Then $W_{i+s} \geq W_i$ for all $i \geq 0$. Hence:*

- (a) *if $W_a = W_{a+1} = \dots = W_{a+s-1} = d$ for some a , then $\mathfrak{r}^a \subseteq A_p$ and $e \leq a$; this applies with $s = E$, since $p \in A_p \cap \mathfrak{r}^E$ has nonzero image in $\text{gr}_E(A)$;*
- (b) *if $W_1 > 0$, then e is the least i with $W_i = d$.*

Proof. The image $V_s = \text{gr}_s(A) \subseteq \bar{R}$ is nonzero and stable under the twisted action, so its projection to every factor k_a is nonzero. If $v_1, \dots, v_r$ is a basis of V_s , the element $v = \sum_k T^{k-1} v_k$ of $\bar{R} \otimes_{\mathbb{F}_p} \mathbb{F}_p(T) = \prod_a k_a(T)$ has every component nonzero, so it is a unit, and multiplication by v embeds $\text{gr}_i(A) \otimes \mathbb{F}_p(T)$ in $\text{gr}_{i+s}(A) \otimes \mathbb{F}_p(T)$; extension of scalars keeps dimensions, so $W_i \leq W_{i+s}$. (a) By induction every layer from a on is full, and $A_p \cap \mathfrak{r}^a$, a closed sublattice with the graded of $\mathfrak{r}^a$, equals $\mathfrak{r}^a$. Since the conductor commutes with $\otimes \mathbb{Z}_p$ and $\mathfrak{r}^a$ is an $\mathcal{O}_p$ -ideal, $\mathfrak{r}^a \subseteq A_p$ gives $\mathfrak{r}^a \subseteq \mathfrak{f}\mathcal{O}_p$, that is, $e \leq a$. (b) Take $s = 1$ in (a): a full layer k gives $e \leq k$; conversely $\mathfrak{r}^e \subseteq A_p$ makes $W_e = d$. $\square$

Since $W_0 = 1 < d$, the layers $0 \leq i < E$ contain at most $E - 1$ full ones, so a computation limited to $i < E$ cannot use (a) with $s = E$; (b) needs only $W_1 > 0$ and one full layer. Several steps can be used at once.

Corollary 7 (certifying by semigroups). *Under the hypotheses of Theorem 5, let Σ be the additive monoid (containing 0) generated by the degrees $s \geq 1$ with $W_s > 0$. If $W_a = d$, then $W_{a+\sigma} = d$ for every $\sigma \in \Sigma$. In particular, if every degree $i \geq a$ is of the form $a' + \sigma$ with $W_{a'} = d$ and $\sigma \in \Sigma$, then $e \leq a$.*

Proof. Apply Proposition 6 once for each generator in a decomposition of σ ; the last statement follows as in Proposition 6(a). $\square$

A second product argument bounds the layers from above.

Proposition 8 (complementary layers). *Under the hypotheses of Theorem 5, if $W_{i+j} < d$ then $W_i + W_j \leq d$. In particular:*

- (a) $W_i + W_{e-1-i} \leq d$ for $0 \leq i < e$, hence $2 \sum_{i < e} W_i \leq ed$;
- (b) if $d/2 < W_1 < d$, then $e = 2$.

For $i + j = e - 1$ the inequality is the diagonal case of [9], Corollary (3.7).

Proof. Extend scalars to $\overline{\mathbb{F}}_p$, so that $\bar{R} \otimes \overline{\mathbb{F}}_p = \overline{\mathbb{F}}_p^d$ with coordinates indexed by geometric points, permuted transitively by $\text{Gal}(K/\mathbb{Q})$, and products of layers are computed coordinatewise. If $V_{i+j} = \text{gr}_{i+j}(A) \otimes \overline{\mathbb{F}}_p$ is a proper subspace, its annihilator in the dual is nonzero and stable under the twisted action; a generic linear combination of the translates of one nonzero annihilating functional is a functional $\lambda(x) = \sum_a \lambda_a x_a$ in the annihilator with every $\lambda_a \neq 0$. The form $B_\lambda(x, y) = \lambda(xy)$ is nondegenerate, and $B_\lambda(V_i, V_j) \subseteq \lambda(V_{i+j}) = 0$, so $W_i + W_j \leq d$. (a) $W_{e-1} < d$: otherwise every layer from $e - 1$ on would be full and $\mathfrak{r}^{e-1} \subseteq A_p$, as in the proof of Proposition 6(a). Summing over i gives the second inequality. (b) $2W_1 > d$ forces $W_2 = d$; by Proposition 6(b), $e \leq 2$, and $e \geq 2$ by Theorem 5(c). $\square$

Proposition 9 (one prime above p). *Under the hypotheses of Theorem 5, suppose that p has a single prime above it in K , of prime residue degree $f = d \geq 3$, and that $2 \leq W_1 < d$. Then $2 \leq e \leq d - 1$. In particular $e = 2$ when $d = 3$.*

Proof. $\bar{R} = \mathbb{F}_{p^d}$. Choose $a, b \in \text{gr}_1(A)$ linearly independent over $\mathbb{F}_p$ and put $t = b/a \notin \mathbb{F}_p$; as d is prime, t generates $\mathbb{F}_{p^d}$, so $1, t, \dots, t^{d-1}$ is a basis. The products of k elements of $\text{gr}_1(A)$ lie in $\text{gr}_k(A)$ and include $a^k, a^{k-1}b, \dots, b^k = a^k(1, t, \dots, t^k)$, which span $\bar{R}$ once $k \geq d - 1$. So every layer $k \geq d - 1$ is full, hence $\mathfrak{r}^{d-1} \subseteq A_p$ (as in the proof of Proposition 6); and $e \geq 2$ by Theorem 5(c). $\square$

Proposition 10 (the Gorenstein criterion, in layers). *Under the hypotheses of Theorem 5, A_p is Gorenstein if and only if $\ell(\mathcal{O}_p/A_p) = \ell(A_p/\mathfrak{f}\mathcal{O}_p)$, that is, if and only if $2 \sum_{i < e} W_i = ed$, that is, if and only if*

$$W_i + W_{e-1-i} = d \quad \text{for every } 0 \leq i < e.$$

When $e = 2$ this reads $W_1 = d - 1$.

Proof. That a Gorenstein ring is characterised by the length equality is classical: Bass proves one direction [7, Cor. (6.5)] and records the converse there after Serre and Grothendieck; see also [16] and [17], Corollary 12.2.4. The symmetric form is the diagonal case of the criterion of Campillo, Delgado and Kiyek [9], (3.6)–(3.7), stated for analytically reduced one-dimensional local Cohen–Macaulay rings with arbitrary residue extensions; we give a direct argument for the case at hand. By Theorem 5(a), $R = A_p$ is local with residue field $\mathbb{F}_p$; it is finite over $\mathbb{Z}_p$, hence complete, reduced of dimension one, with finite normalisation $\mathcal{O}_p$ and conductor $\mathfrak{f}\mathcal{O}_p = \mathfrak{f} \otimes \mathbb{Z}_p$ (flat base change). Every composition factor of a finite R -module is $\mathbb{F}_p$, so its length is $\log_p$ of its cardinality; the layers are killed by p , and their lengths are $\mathbb{F}_p$ -dimensions. So the first

two forms of the condition agree by Theorem 5(d), and the third is the case of equality in Proposition 8(a). If R is Gorenstein the equality holds by [7, Cor. (6.5)], or [17], Theorem 12.2.2, which makes no assumption on the residue field. The converse is [17], Corollary 12.2.4, for a one-dimensional analytically unramified local ring with infinite residue field and a canonical module contained in the ring; we reach it through $S = R[T]_{\mathfrak{m}_R[T]}$. The map $R \rightarrow S$ is flat and local with closed fibre $\mathbb{F}_p(T)$. The ring S is a localisation of a finitely generated algebra over a complete local ring, hence excellent; being excellent and reduced, it is analytically unramified. Its normalisation is the corresponding localisation of $\mathcal{O}_p[T]$, which is normal and finite over $R[T]$. The conductor is $\text{Ann}_R(\mathcal{O}_p/R)$, and annihilators of finitely generated modules commute with flat base change, so the conductor of S is $\mathfrak{f}S$. Lengths are preserved, since each composition factor $\mathbb{F}_p$ becomes $\mathbb{F}_p(T)$. Finally S is Cohen–Macaulay (reduced of dimension one), a quotient of a regular local ring, and generically Gorenstein, so it has a canonical module isomorphic to an ideal [8, Thm. 3.3.6, Prop. 3.3.18(b)]. Hence the equality for R gives the equality for S , so S is Gorenstein by [17], Corollary 12.2.4, and so is R , because a flat local homomorphism of noetherian local rings with Gorenstein target has Gorenstein source [26, Tag 0BJL]. $\square$

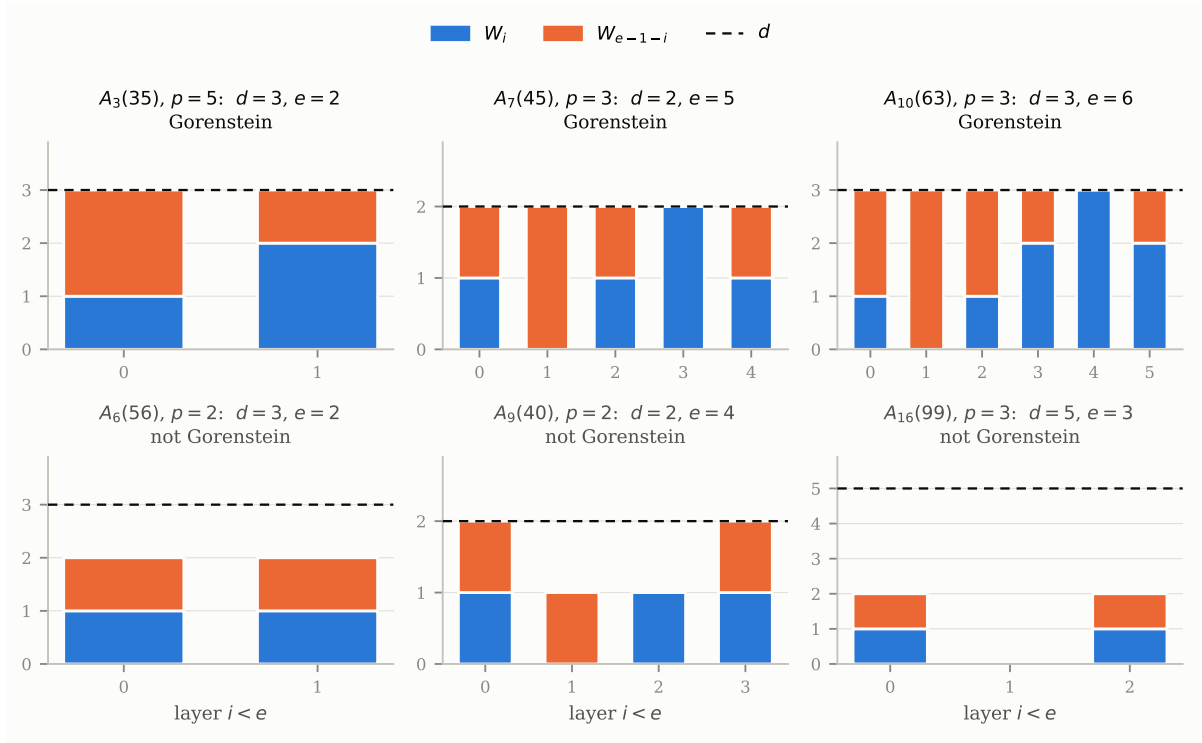


Figure 4: Proposition 10 on six lattice computations. Column $i < e$ stacks W_i and its mirror W_{e-1-i} ; by Proposition 8 no column passes d , and the local ring is Gorenstein exactly when every column reaches it. Top: $A_3(35)$, $A_7(45)$, $A_{10}(63)$; bottom: $A_6(56)$, $A_9(40)$, $A_{16}(99)$. In each panel the symmetry agrees with $v_p([\mathcal{O} : A]) = v_p([A : \mathfrak{f}])$ computed on the lattice.

The Gorenstein property is the case $\text{type}(A_p) = 1$ of a finer invariant. When the conductor is the square of the radical, the type is read off the first layer and its multiplier ring.

Proposition 11 (the type when $e = 2$). *Under the hypotheses of Theorem 5, suppose $e = 2$. Let $V = \text{gr}_1(A) \subseteq \bar{R}$ and $\kappa = \dim_{\mathbb{F}_p}\{a \in \bar{R} : aV \subseteq V\}$. Then the Cohen–Macaulay type of A_p is*

$$\text{type}(A_p) = d - W_1 + \kappa - 1,$$

$d - W_1 \geq \kappa$, and A_p is almost Gorenstein, $\ell(\mathcal{O}_p/A_p) = \ell(A_p/\mathfrak{f}\mathcal{O}_p) + \text{type}(A_p) - 1$, if and only if $d - W_1 = \kappa$.

Proof. Let $\mathfrak{m} = A_p \cap \mathfrak{r}$, the maximal ideal of A_p . Since $A_p \neq \mathcal{O}_p$, $\mathfrak{m}$ is not invertible [23, Lemma 2.14], and by [23], Proposition 3.5, $\text{type}(A_p) + 1 = \dim_{\mathbb{F}_p}(\mathfrak{m} : \mathfrak{m})/\mathfrak{m}$, and $(\mathfrak{m} : \mathfrak{m})$, a ring finite over A_p , lies in $\mathcal{O}_p$. As $e = 2$, $\mathfrak{r}^2 \subseteq A_p$ and $\mathfrak{m}/\mathfrak{r}^2$ is the layer V . For $x \in \mathcal{O}_p$, $x\mathfrak{r}^2 \subseteq \mathfrak{r}^2$, and $x\mathfrak{m} \subseteq \mathfrak{m}$ if and only if the residue $\bar{x}$ satisfies $\bar{x}V \subseteq V$. Hence $(\mathfrak{m} : \mathfrak{m}) = \{x \in \mathcal{O}_p : \bar{x}V \subseteq V\}$ contains $\mathfrak{r}$, and $\dim(\mathfrak{m} : \mathfrak{m})/\mathfrak{m} = \kappa + (d - W_1)$. By Theorem 5(d), $\ell(\mathcal{O}_p/A_p) = 2d - 1 - W_1$ and $\ell(A_p/\mathfrak{f}\mathcal{O}_p) = 1 + W_1$, so the almost Gorenstein equality reads $d - W_1 = \kappa$. The inequality is [17], Theorem 12.2.3, applied to $S = A_p[T]_{\mathfrak{m}_{A_p}[T]}$ as in the proof of Proposition 10; lengths and type do not change under that flat local extension with field fibre, because Ext commutes with flat base change. $\square$

The definition of almost Gorenstein rings is that of Barucci and Fröberg [2], in the length form used by Marseglia [23], §7.2. On the 403 pairs (order, prime) with $q \leq 250$ in which the computed layers certify $e = 2$, the type computed from the colon ideal modulo $\mathfrak{r}^2$ agrees with the formula, including 19 with $\kappa > 1$, 9 of them also by a colon on the lattice; so do the 63 lattice pairs with $e = 2$ (*measured*, §9).

6 Why do class numbers appear?

Theorem 5 reduces the first layer, and with it the criterion for the conductor at p to be the radical, to the rank of a single matrix built from S . Where have such matrices been seen before?

Look at S on the families of Table 1. If $m = Nq'$ or $m = Nq' - 1$, then $S(c) = N(2c - q')$ for $1 \leq c \leq q' - 1$: up to the factor $2Nq'$, the sawtooth $B_1(c/q')$, whose coefficients are those of Stickelberger's element. If $2m + 1 = Nq'$ with q' odd, then $S(c) = N\hat{c}$, with $\hat{c} \in (-q'/2, q'/2)$ the signed residue: the same sawtooth shifted by half a period (Figure 5). For even $q' = 2r$, the two remaining stable families of Table 1, $m = Br$ and $m = Br - 1$ with B odd, give the shifted sawtooth of period q' as well: $S(c) = B\hat{s}_{q'}(c)$, where $\hat{s}_{q'}(c) = \hat{c}$ is the signed residue modulo q' and $\hat{s}_{q'}(r) = 0$.

Only these two functions occur, and one line says why. A sum is its count times its mean: writing $C(c)$ for the number of $j \in [-m, m]$ with $j \equiv c \pmod{q'}$ and $\mu(c)$ for the midpoint of that arithmetic progression, $S(c) = C(c)\mu(c)$ whenever it is nonempty. What the family fixes is the midpoint. For $m \equiv 0, -1 \pmod{q'}$ every class $c \neq 0$ has $C(c) = 2N$ and $\mu(c) = c - q'/2$; for $2m + 1 = Nq'$ every class has $C(c) = N$ and $\mu(c) = \hat{c}$; on the half-period families $C(c) = B$ and $\mu(c) = \hat{c}$ away from the two classes 0 and $q'/2$ fixed by $c \mapsto -c$, where S vanishes anyway. The sawtooth and its shift are the two midpoints, and the multiplicity is the count.

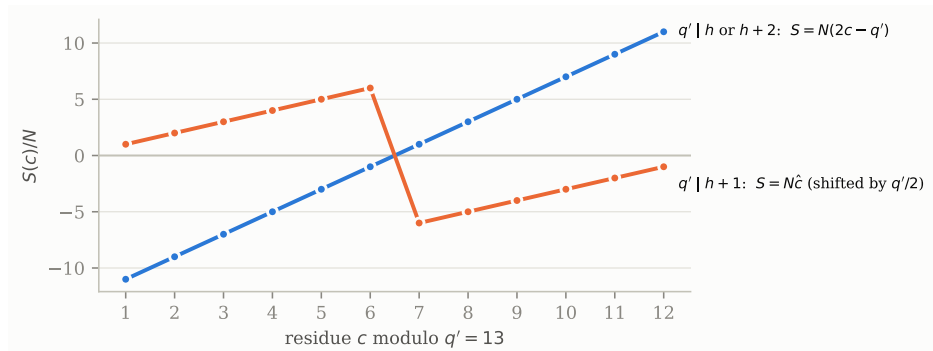


Figure 5: The first-moment function S for $q' = 13$ on the two kinds of stable segment.

For the first two families the matrix of Theorem 5(b) is, up to the factor N , the *moment matrix*

$$M_n = [s_n(u^{-1}c)], \quad s_n(0) = 0, \quad s_n(c) = 2c - n \quad (1 \leq c < n),$$

with rows u running over representatives of $(\mathbb{Z}/n)^\times / \pm 1$ and columns over *all* residues $1 \leq c < n$, units or not. Its maximal minors are governed by Sinnott's index of the Stickelberger ideal. The two kinds of segment lead to two different pieces of arithmetic. The answers, stated together:

Theorem 12 (class numbers). *Let p be odd and $p \mid q$.*

- (a) *If $q' \notin \{1, 2, 3, 4, 6\}$, $m \equiv 0$ or $-1 \pmod{q'}$ and $p \nmid N := \lceil m/q' \rceil$, then $d - W_1$ is the number of elementary divisors of $M_{q'}$ divisible by p . The same holds on the half-period families: for $q' \geq 8$ even, $r = q'/2$, $m \equiv r$ or $r - 1 \pmod{q'}$ and $p \nmid B := \lceil m/r \rceil$. In both cases $\mathfrak{f}\mathcal{O}_p = \mathfrak{r}$ if and only if $p \nmid h^-(\mathbb{Q}(\zeta_{q'}))$.*
- (b) *If $q' \geq 5$ is prime and $2m + 1 = Nq'$ with $p \nmid N(q' - 1)$, then*

$$W_1 = \#\{\chi \text{ odd mod } q' : (\bar{\chi}(2) - 1) B_{1,\chi} \not\equiv 0 \pmod{\mathfrak{p}}\}$$

for any prime $\mathfrak{p}$ above p in $\mathbb{Q}(\mu_{q'-1})$.

6.1 The sawtooth and Stickelberger's ideal

Proposition 13 (Sinnott's index, on the moment matrix). *Let $n \geq 3$, $n \not\equiv 2 \pmod{4}$, $d = \varphi(n)/2$, and let Δ_n be the greatest common divisor of the $d \times d$ minors of M_n . Then*

$$\Delta_n = 2^a h^-(\mathbb{Q}(\zeta_n)) \frac{(2n)^d}{w_n}, \quad (5)$$

where w_n is the number of roots of unity in $\mathbb{Q}(\zeta_n)$, $a = 0$ if n is a prime power and $a = 2^{t-2} - 1$ if n has $t \geq 2$ prime factors. In particular, for an odd prime $p \nmid n$, $v_p(\Delta_n) = v_p(h^-)$, and the rank of M_n modulo p is d minus the number of elementary divisors of M_n divisible by p .

Proof. This is Sinnott's index formula [25, 27] read through the generators of Bernard and Kučera [3]. Let $G = (\mathbb{Z}/n)^\times$ and, for $0 < c < n$, $\omega_n(c) = \sum_{u \in G} (\langle -cu/n \rangle - \frac{1}{2}) \sigma_u^{-1}$, with $\langle x \rangle$ the fractional part. Replacing u by u^{-1} and using $\langle -x \rangle - \frac{1}{2} = -(\langle x \rangle - \frac{1}{2})$ for $x \notin \mathbb{Z}$,

$$\sum_{u \in G} s_n(u^{-1}c) \sigma_u = -2n \omega_n(c),$$

for units and non-units alike. Each $\omega_n(c)$ is odd, so the columns of M_n are the coordinates of $-2n \omega_n(c)$ in the basis $\sigma_u - \sigma_{-u}$ of $R^- = \{\alpha \in \mathbb{Z}[G] : (1 + \sigma_{-1})\alpha = 0\}$, and $\Delta_n = [R^- : 2n\Omega]$, where Ω is the $\mathbb{Z}$ -span of the $\omega_n(c)$. By [3, Lemma 1.2 and (7)], Sinnott's group S' is generated by Ω and $\frac{1}{2}N_G$, $N_G = \sum_{u \in G} \sigma_u$; the Stickelberger ideal is $S = S' \cap \mathbb{Z}[G]$ [3, §1]; since Ω is odd and N_G even, $S' \cap R^- \otimes \mathbb{Q} = \Omega$ and $S^- = S \cap \Omega$; by [3, Lemma 1.1] ([27, Proposition 2.1]), $[S' : S] = w_n$; and by [3, (26)], from [25], $[R^- : S^-] = 2^a h^-$ with $S^- = S \cap R^-$. An element $\alpha \in S$ with $(1 + \sigma_{-1})\alpha = N_G$ [3, Proposition 3.1] puts $\frac{1}{2}N_G$ in $S + \Omega$, so $S' = S + \Omega$ and $[\Omega : S^-] = [S' : S] = w_n$. As Ω has rank d and $2n \omega_n(c) = 2n \theta_n(c) - n N_G \in S^-$, with $\theta_n(c) = \sum_{u \in G} \langle -cu/n \rangle \sigma_u^{-1}$, $[S^- : 2n\Omega] = (2n)^d / w_n$, and $\Delta_n = [R^- : S^-] [S^- : 2n\Omega]$. $\square$

The rectangular matrix itself is not new: Kuribayashi [22], Proposition, proves that it has rank $\varphi(n)/2$ over $\mathbb{Q}$ for every $n > 2$, in connection with a conjecture of Breuer [5], Conjecture C.12; (5) refines that rank to the product of the elementary divisors, which is what the greatest common divisor of the maximal minors is; how the p -part is distributed among them it does not say (§8). The non-unit columns cannot be dropped: by [3, (6)] they are the corestrictions of the Stickelberger elements of the lower levels $n/\gcd(c, n)$, and without them the maximal minors may all vanish (Remark 15). *Measured*: on the 66 values $n < 90$, $n \not\equiv 2 \pmod{4}$, the elementary divisors computed in Sage give $v_p(\Delta_n) = v_p(h^-)$ for exact h^- in all 1505 pairs with p odd, $p \leq 97$, $p \nmid n$, including the 40 pairs with $p \mid \varphi(n)$, where the group algebra is not semisimple; the whole of (5), powers of 2 and of n included, holds on all 66 (§9).

Sinnott's normalisation excludes $n \equiv 2 \pmod{4}$, where $\mathbb{Q}(\zeta_n) = \mathbb{Q}(\zeta_{n/2})$. At odd primes the moment matrix does not need the exclusion: doubling the period doubles its column lattice, although the power of 2 in (5) changes.

Lemma 14 (half period, and doubling). *Let $n \geq 4$ be even, let $\widehat{M}_n = [\hat{s}_n(u^{-1}c)]$, $0 \leq c < n$, and let L_n and L'_n inside $\mathbb{Z}^{\varphi(n)/2}$ be the spans of the columns of M_n and of $\widehat{M}_n$.*

(i) *For every c ,*

$$\hat{s}_n(c) = \frac{1}{2} s_n(c + n/2), \quad \text{hence} \quad L'_n = \frac{1}{2} L_n.$$

(ii) *If $n = 2r$ with $r \geq 3$ odd, and the rows of M_r and M_{2r} are indexed by the same odd representatives u of $(\mathbb{Z}/r)^\times / \pm 1 = (\mathbb{Z}/2r)^\times / \pm 1$, then*

$$L_{2r} = 2L_r, \quad \text{hence} \quad L'_{2r} = L_r.$$

So the elementary divisors of $\widehat{M}_n$ are half those of M_n , those of M_{2r} are twice those of M_r , $\Delta_{2r} = 2^d \Delta_r$ with $d = \varphi(r)/2$, and for odd p the matrices M_n and $\widehat{M}_n$ — and, for $n = 2r$, also M_r — have the same rank modulo p .

Proof. (i) For $0 < c < n/2$ one has $c + n/2 \in (n/2, n)$ and $s_n(c + n/2) = 2c = 2\hat{s}_n(c)$; for $n/2 < c < n$ one has $c + n/2 \equiv c - n/2 \in (0, n/2)$ and $s_n(c - n/2) = 2c - 2n = 2\hat{s}_n(c)$; and both sides vanish at $c = 0$ and at $c = n/2$, by the conventions $s_n(0) = 0$ and $\hat{s}_n(n/2) = 0$. A unit u modulo the even n is odd, so $u^{-1}(c + n/2) = u^{-1}c + n/2$ and the identity holds entry by entry: the column c of $\widehat{M}_n$ is half the column $c + n/2$ of M_n . Adjoining to M_n the zero column $c = 0$, which does not change L_n , the map $c \mapsto c + n/2$ permutes the columns, so $L'_n = \frac{1}{2} L_n$; the halves are integral because n is even, so every entry of M_n is.

(ii) For c modulo r let $\nu(c)$ be the odd one of its two lifts modulo $2r$, and $c^* \in [0, r)$ the residue of $2^{-1}c$, so that $c^* = (c + r)/2$ for odd $c \in [0, r)$ and $c^* = c/2$ for even c . Checking these two cases,

$$s_{2r}(2c) = 2s_r(c), \quad s_{2r}(\nu(c)) = 2(s_r(c) - s_r(c^*)).$$

As u is odd, $u^{-1} \cdot 2c = 2(u^{-1}c)$ and $u^{-1}\nu(c) = \nu(u^{-1}c)$ modulo $2r$, and $(u^{-1}c)^* = u^{-1}c^*$ modulo r , so these identities hold column by column. The even columns of M_{2r} are twice the columns of M_r , all of them as c runs over $\mathbb{Z}/r$, and its odd columns are twice differences of them, so $L_{2r} = 2L_r$; with (i), $L'_{2r} = \frac{1}{2} L_{2r} = L_r$. $\square$

Measured for odd $r \leq 45$: the identities of (ii), the lattices, the elementary divisors and Δ_{2r} ; and, end to end on the layers, in 304 cases with $q' = 2r$, $r \in \{5, 7, 11, 13, 23, 29, 31\}$ and $p \in \{3, 5, 7\}$, in all four families (first layer only), including $p = 3$ with $r = 23, 31$, where $p \mid h^-$. For (i): the shift identity, $L'_n = \frac{1}{2} L_n$, the elementary divisors and the equality of ranks at $p \leq 13$, on the 34 even $n \leq 70$; and, end to end on the layers, in 156 cases with $4 \mid q'$, $q' \leq 88$, on the two

half-period families, among them the three pairs with $p \mid h^-$, namely $(q', p) = (52, 3), (64, 17)$ and $(88, 5)$, where $W_1 = d - 1$ (§9).

One identity, two regimes. Part (i) and the Euler factor of §6.2 are the same identity. For every n and every c ,

$$2 \hat{s}_n(c) = s_n(2c) - s_n(c),$$

the distribution relation of B_1 read on residues. What changes is whether 2 is invertible modulo n . For n odd, $c \mapsto 2c$ permutes $\mathbb{Z}/n$ and the identity becomes $[\hat{s}_n(u^{-1}c)] = \frac{1}{2}(P - I)M_n$ with P the signed permutation of $u \mapsto u/2$; on characters it is the Euler factor $\bar{\chi}(2) - 1$, which vanishes at the odd characters trivial on $\langle 2 \rangle$, and that vanishing is the defect of Corollary 17. For n even, $c \mapsto 2c$ is not injective, but $s_n(2c) - s_n(c) = s_n(c + n/2)$: the identity collapses to a translation by half a period, which permutes the columns and cannot lose rank. So the factor at 2 belongs to odd q' alone, and the same $\hat{s}$ serves both regimes. Nothing in the argument of Corollary 17 is special to 2: for any ℓ prime to n the matrix of $\frac{1}{2}(s_n(\ell c) - s_n(c))$ has rank $\varphi(n)/2 - \delta_\ell$ over $\mathbb{Q}$, and the same rank modulo every odd prime $p \nmid n h^-$, with $\delta_\ell = 0$ if $-1 \in \langle \ell \rangle$ and $\delta_\ell = \varphi(n)/(2 \text{ord}_n(\ell))$ otherwise (*measured* on the 52 pairs (n, ℓ) with $n \leq 35$ odd and $\ell \leq 7$, over $\mathbb{Q}$ and modulo the odd primes $p \leq 13$ with $p \nmid n h^-$). The parity of p is not cosmetic: for $n = 5$ and $\ell = 4$ the matrix has rows $(3, 1, -1, -3)$ and $(-1, 3, -3, 1)$, all entries odd, so its rank modulo 2 is 1, while $-1 \in \langle 4 \rangle$ and $\delta_4 = 0$. Only $\ell = 2$ occurs here, because the only shift a symmetric segment can produce is half a period.

Proof of Theorem 12(a). Here $A \neq \mathbb{Z}$, and $E \geq 2$ since p is odd. In both families $S(c) = N s_{q'}(c)$ for $0 \leq c < q'$, so by Theorem 5(b) W_1 is the rank modulo p of $NM_{q'}$, which is that of $M_{q'}$ because $p \nmid N$; Proposition 13 (with $p \nmid q'$) and Theorem 5(c) conclude. For $q' = 2r$ with r odd, Proposition 13 does not apply at q' , but Lemma 14(ii) gives $M_{q'}$ and M_r the same number of elementary divisors divisible by p , and $\mathbb{Q}(\zeta_{q'}) = \mathbb{Q}(\zeta_r)$, so it applies at r instead. On the half-period families, for any even q' , counting as in the proof of Theorem 4(c) gives $S = B \hat{s}_{q'}$ with $B = \lceil m/r \rceil$ odd: writing $m = kq' + r$ or $m = kq' + r - 1$, the k complete periods contribute $k(2c - q')$ to $S(c)$ and the remaining segment contributes $kq' + c$ for $0 < c < r$ and $-(kq' + q' - c)$ for $r < c < q'$, so $S(c) = (2k + 1)\hat{c}$ and $B = 2k + 1$. Hence W_1 is the rank modulo p of $\widehat{M}_{q'}$, which is that of $M_{q'}$ by Lemma 14(i). For q' prime the block of $M_{q'}$ on $u, c < q'/2$ is 2 times Maillet's reduced matrix up to transposition, and this is the evaluation of Carlitz and Olson [10], (1.7) and (2.5). $\square$

Remark 15 (squares and rectangles). For q' prime, Proposition 13 is Carlitz and Olson's evaluation of Maillet's determinant. For composite q' the classical determinants are square. The matrix on units, evaluated by Tateyama and by Wang [21], (2), vanishes as soon as some prime divisor of q' splits in $\mathbb{Q}(\zeta_{q'})/\mathbb{Q}(\zeta_{q'})^+$; Kučera's determinants [21], Theorem 1, combine Stickelberger elements of all levels $n \mid q'$ with rational weights; their value carries Euler factors $\prod(1 - \bar{\chi}(\ell))$, which vanish for some choices of weights, and Kučera chooses weights that give nonvanishing determinants for every conductor. Girstmair's index formula [14] uses the conjugates of a single cotangent number inside the field, with explicit ramification factors. The rectangular matrix $M_{q'}$, whose rank over $\mathbb{Q}$ is Kuribayashi's [22], spans $2q'\Omega$, of index $(2q')^d/w_{q'}$ in the minus part of the Stickelberger ideal (for $q' \equiv 2 \pmod{4}$, read at $r = q'/2$ through Lemma 14(ii)), which is a unit at every odd $p \nmid q'$; no choice of weights is needed. Figure 7 shows it at work on the non-prime values: an earlier measurement over 984 pairs (q', p) with $3 \leq q' < 90$, $q' \not\equiv 2 \pmod{4}$ and odd $p \leq 60$, $p \nmid q'$, is now a consequence of Proposition 13.

A simple divisor of the class number fixes the whole local picture.

	p=3	p=5	p=7	p=11	p=13
$q'=5 \quad h^- = 1$		—			
$q'=7 \quad h^- = 1$			—		
$q'=11 \quad h^- = 1$				—	
$q'=13 \quad h^- = 1$					—
$q'=17 \quad h^- = 1$					
$q'=19 \quad h^- = 1$					
$q'=23 \quad h^- = 3$	1				
$q'=29 \quad h^- = 2^3$					
$q'=31 \quad h^- = 3^2$	1				
$q'=37 \quad h^- = 37$					
$q'=41 \quad h^- = 11^2$				2	
$q'=43 \quad h^- = 211$					
$q'=47 \quad h^- = 5 \cdot 139$		1			
$q'=53 \quad h^- = 4889$					
$q'=59 \quad h^- = 3 \cdot 59 \cdot 233$	1				
$q'=61 \quad h^- = 41 \cdot 1861$					
$q'=67 \quad h^- = 67 \cdot 12739$					
$q'=71 \quad h^- = 7^2 \cdot 79241$			1		
$q'=73 \quad h^- = 89 \cdot 134353$					
$q'=79 \quad h^- = 5 \cdot 53 \cdot 377911$		1			
$q'=83 \quad h^- = 3 \cdot 279405653$	1				
$q'=89 \quad h^- = 113 \cdot 118401449$					

Figure 6: The defect $\varphi(q')/2 - W_1$ for $S = 2c - q'$, over primes $q' \leq 89$ and $p \leq 13$ (a dash where $p = q'$), with $h^-(\mathbb{Q}(\zeta_{q'}))$ beside each row. The nonzero cells are exactly the pairs with $p \mid h^-$.

Corollary 16 (a simple divisor of h^-). *In Theorem 12(a), suppose that $v_p(h^-(\mathbb{Q}(\zeta_{q'}))) = 1$. Then $W_1 = d - 1$,*

$$\mathfrak{f} \mathcal{O}_p = \mathfrak{r}^2, \quad v_p([\mathcal{O} : A]) = v_p([A : \mathfrak{f}]) = d,$$

and A_p is Gorenstein.

Proof. By Proposition 13, and Lemma 14(ii) when $q' \equiv 2 \pmod{4}$, exactly one elementary divisor of $M_{q'}$ is divisible by p , so $W_1 = d - 1$ by Theorem 12(a). Since $p \mid h^-$ forces $\varphi(q') > 20$ [29, Ch. 11], so $d \geq 11$, $d/2 < d - 1 < d$, and Proposition 8(b) gives $e = 2$; Theorem 5(d) gives the two lengths, and Proposition 10 with $W_1 = d - 1$ gives the Gorenstein property. $\square$

The hypothesis is sufficient, not necessary: a larger power of p in h^- can still leave a defect of one (*measured*: $(q', p) = (31, 3)$ and $(71, 7)$, with $v_p(h^-) = 2$ and rank $d - 1$). **A second example.** The running example $A_3(35)$ owes its defect to the factor at 2, with $h^- = 1$. The order $A_{22}(69)$ owes it to the class number: $q = 69$, $m = 22 \equiv -1 \pmod{23}$, $p = 3$, $N = 1$, $d = 11$ and $h^-(\mathbb{Q}(\zeta_{23})) = 3$. Corollary 16 gives $W_1 = 10$, $\mathfrak{f} \mathcal{O}_3 = \mathfrak{r}^2 = 3\mathcal{O}_3$, $v_3([\mathcal{O} : A]) = v_3([A : \mathfrak{f}]) = 11$, and a Gorenstein ring $A_{22}(69) \otimes \mathbb{Z}_3$. A lattice computation, independent of these arguments, returns $[\mathcal{O} : A_{22}(69)] = 3^{11}$, $\mathfrak{f} = 3\mathcal{O}$, layers $(W_0, W_1, W_2, W_3) = (1, 10, 11, 11)$ and Cohen–Macaulay type 1 (§9). A class number can also leave a ring that is not almost Gorenstein. For $A_{40}(451)$ at $p = 11$, with $h^-(\mathbb{Q}(\zeta_{41})) = 11^2$, the layers give $W_1 = 18$ and $W_2 = d = 20$, so $e = 2$; the multiplier ring of the first layer is $\mathbb{F}_{11}$, $\kappa = 1$, and Proposition 11 gives type 2, while $\ell(\mathcal{O}_{11}/A_{11}) - \ell(A_{11}/\mathfrak{f}\mathcal{O}_{11}) = 21 - 19 = 2 \neq$ type -1 (*computed* from the layers, not from a lattice).

		p=3	p=5	p=7	p=11	p=13	p=17
$q'=8$	$h^-=1$						
$q'=9$	$h^-=1$	—					
$q'=12$	$h^-=1$	—					
$q'=15$	$h^-=1$	—	—				
$q'=16$	$h^-=1$						
$\dagger q'=20$	$h^-=1$		—				
$\dagger q'=21$	$h^-=1$	—		—			
$\dagger q'=24$	$h^-=1$	—					
$q'=25$	$h^-=1$		—				
$q'=27$	$h^-=1$	—					
$\dagger q'=28$	$h^-=1$			—			
$q'=32$	$h^-=1$						
$\dagger q'=33$	$h^-=1$	—			—		
$q'=35$	$h^-=1$		—	—			
$q'=36$	$h^-=1$	—					
$\dagger q'=39$	$h^-=2$	—				—	
$\dagger q'=40$	$h^-=1$		—				
$q'=44$	$h^-=1$				—		
$q'=45$	$h^-=1$	—	—				
$\dagger q'=48$	$h^-=1$	—					
$q'=49$	$h^-=43$			—			
$q'=51$	$h^-=5$	—	1				—
$\dagger q'=52$	$h^-=3$	1				—	
$\dagger q'=55$	$h^-=2 \cdot 5$		—		—		
$\dagger q'=56$	$h^-=2$			—			
$\dagger q'=57$	$h^-=3^2$	—					
$\dagger q'=60$	$h^-=1$	—	—				
$\dagger q'=63$	$h^-=7$	—		—			
$q'=64$	$h^-=17$						1
$q'=65$	$h^-=2^6$		—			—	
$\dagger q'=68$	$h^-=2^3$						—
$\dagger q'=69$	$h^-=3 \cdot 23$	—					
$\dagger q'=72$	$h^-=3$	—					
$q'=75$	$h^-=11$	—	—		1		
$q'=76$	$h^-=19$						
$\dagger q'=77$	$h^-=2^8 \cdot 5$		1	—	—		
$\dagger q'=80$	$h^-=5$		—				
$q'=81$	$h^-=2593$	—					
$\dagger q'=84$	$h^-=1$	—		—			
$q'=85$	$h^-=5 \cdot 17 \cdot 73$		—				—
$q'=87$	$h^-=2^9 \cdot 3$	—					
$\dagger q'=88$	$h^-=5 \cdot 11$		1		—		

Figure 7: Theorem 12(a) beyond prime q' : the rank defect of $M_{q'}$ modulo p , for the 42 non-prime q' with $8 \leq q' < 90$ and $q' \not\equiv 2 \pmod{4}$, and $p \leq 17$ (a dash where $p \mid q'$), with the exact $h^-(\mathbb{Q}(\zeta_{q'}))$ beside each row. The nonzero cells are exactly the pairs with $p \mid h^-$. A dagger marks the 21 values at which the square matrix on units alone is singular over $\mathbb{Q}$.

6.2 The shifted sawtooth and the factor at 2

Proof of Theorem 12(b). The multiplication formula $B_1(2x) = B_1(x) + B_1(x + \frac{1}{2})$ gives $\hat{c}/q' = B_1(\{c/q' + \frac{1}{2}\}) = B_1(\{2c/q'\}) - B_1(\{c/q'\})$; summing against an odd character, $\sum_c \chi(c)\hat{c} = q'(\bar{\chi}(2) - 1)B_{1,\chi}$. This identity, with the same Euler factor, is in Kanemitsu and Kuzumaki [19], proof of Theorem 1 ($t = 2$, odd modulus), and in Kučera [21], Example 2, where it gives the determinant of the square matrix of signed least residues; for the half sums $\sum_{a < q'/2} \chi(a)$ see [28], (6). When $p \nmid q' - 1$ the algebra $\mathbb{F}_p[(\mathbb{Z}/q')^\times]$ is semisimple, and the dimension of the cyclic module generated by the odd function S is the number of odd characters at which its Fourier transform is nonzero modulo $\mathfrak{p}$; the count does not depend on $\mathfrak{p}$, since Galois permutes the characters. $\square$

When the class number plays no role, the Euler factor makes everything explicit, with no condition on $\varphi(q')$, and for composite q' as well; what decides is the subgroup $\langle 2 \rangle$ of $(\mathbb{Z}/q')^\times$. When no prime divisor of q' splits in $\mathbb{Q}(\zeta_{q'})/\mathbb{Q}(\zeta_{q'})^+$ (for instance when q' is a prime power),

part (a) below also follows from the determinant of Kanemitsu–Kuzumaki [19] and Kučera [21], (7) and Example 2, which is then nonzero exactly when $-1 \in \langle 2 \rangle$; part (a) for the remaining composite q' , and the rank, lengths and type in part (b), we have not found.

Corollary 17 (the factor at 2). *Let p be odd, $p \mid q$, $q' \geq 5$ odd, $2m + 1 = Nq'$ and $p \nmid N h^-(\mathbb{Q}(\zeta_{q'}))$; let t be the order of 2 in $(\mathbb{Z}/q')^\times$ and $d = \varphi(q')/2$.*

(a) *If $-1 \in \langle 2 \rangle$, then $W_1 = d$, $\mathfrak{f} \mathcal{O}_p = \mathfrak{r}$, $v_p([\mathcal{O} : A]) = d - 1$, and A_p has Cohen–Macaulay type $d - 1$.*

(b) *If $-1 \notin \langle 2 \rangle$, put $\delta = d/t$. Then*

$$\boxed{\begin{array}{ll} W_1 = d - \delta, & \mathfrak{f} \mathcal{O}_p = \mathfrak{r}^2, \\ v_p([\mathcal{O} : A]) = d + \delta - 1, & v_p([A : \mathfrak{f}]) = d - \delta + 1, \end{array}} \quad (6)$$

and A_p is almost Gorenstein of type $2\delta - 1$; it is Gorenstein exactly when $\delta = 1$.

Proof. For every residue c modulo q' one has $\hat{c} = \frac{1}{2}(s_{q'}(2c) - s_{q'}(c))$. Hence, with rows u over representatives of $(\mathbb{Z}/q')^\times / \pm 1$ and columns over all residues,

$$[S(u^{-1}c)] = \frac{N}{2} (P - I) M_{q'},$$

where P is the signed permutation matrix of $u \mapsto u/2$. As $p \nmid h^-$, Proposition 13 makes $M_{q'}$ of rank d modulo p , so its columns span $\mathbb{F}_p^d$, and the column space of $[S(u^{-1}c)]$ is the image of $P - I$. Multiplication by 2 splits $(\mathbb{Z}/q')^\times / \pm 1$ into orbits. If $-1 \in \langle 2 \rangle$ they have length $t/2$ and P has monodromy -1 along each ($2^{t/2} = -1$), so $P - I$ is invertible. If $-1 \notin \langle 2 \rangle$, then $2^k u \neq \pm u$ for $0 < k < t$, so there are δ orbits of length t , with $t \geq 3$ because $q' \nmid 2^2 - 1$; P has monodromy $+1$, and the image of $P - I$ is the direct sum, over the orbits, of the hyperplanes $\sum \epsilon_u x_u = 0$ with all $\epsilon_u = \pm 1$. By Theorem 5(b) this gives W_1 , and (a) follows from Theorem 5(c) and (d); there $A_p = \mathbb{Z}_p + \mathfrak{r}$ by Theorem 5(a), its maximal ideal is $\mathfrak{r}$, which is not invertible since $A_p \neq \mathcal{O}_p$, and [23], Proposition 3.5, gives $\text{type}(A_p) + 1 = \dim_{\mathbb{F}_p} \mathcal{O}_p / \mathfrak{r} = d$. In (b), $t \geq 3$ gives $d/2 < W_1 < d$, so $e = 2$ by Proposition 8(b), and Theorem 5(d) gives the lengths. By (4), $V \otimes \overline{\mathbb{F}}_p$ is this direct sum of hyperplanes rescaled by the units U_u^{-1} coordinatewise; a diagonal rescaling does not change the multiplier ring, and the vectors a with $aH \subseteq H$, for a hyperplane H of $\overline{\mathbb{F}}_p^t$ ($t \geq 2$) with all coefficients nonzero, are the constants. So $\kappa = \delta$, and Proposition 11 gives type $2\delta - 1$ and the almost Gorenstein equality. $\square$

The running example, concluded. Take $q' = 7$, $m = 3$, $p = 5$, so $q = 35$ and $h^-(\mathbb{Q}(\zeta_7)) = 1$. Here $2m + 1 = 7$, and the singular matrix of §5 is explained by the odd character $\chi = (\frac{\cdot}{7})$, for which $\chi(2) = 1$ kills the Euler factor. The order of 2 mod 7 is $t = 3$, so $-1 \notin \langle 2 \rangle$, $\delta = 1$, and Corollary 17(b) gives $W_1 = 2$ and $e = 2$. The order of 5 in $(\mathbb{Z}/7)^\times / \pm 1$ is $3 = d$, so 5 has a single prime above it, with residue field $\mathbb{F}_{125}$. By Theorem 5(d),

$$\mathfrak{f} \mathcal{O}_5 = \mathfrak{r}^2, \quad v_5([A : \mathfrak{f}]) = 1 + 2 = 3, \quad v_5([\mathcal{O} : A]) = 6 - 1 - 2 = 3,$$

so $[\mathcal{O} : A_3(35)] = 125$ and $[\mathcal{O} : \mathfrak{f}] = 5^6$, and by Proposition 10 the ring $A_3(35) \otimes \mathbb{Z}_5$ is Gorenstein. (The lattice computation of §9 gives the same numbers.)

A larger δ . For $A_{15}(155)$ at $p = 5$: $q' = 31$, $d = 15$, $t = 5$, $\delta = 3$ and $h^-(\mathbb{Q}(\zeta_{31})) = 9$. Here $5 \mid q' - 1$, outside Theorem 12(b), but inside Corollary 17: $W_1 = 12$, $\mathfrak{f} \mathcal{O}_5 = \mathfrak{r}^2$, $v_5([\mathcal{O} : A]) = 17$, $v_5([A : \mathfrak{f}]) = 13$, and type 5, so $A_{15}(155) \otimes \mathbb{Z}_5$ is almost Gorenstein but not Gorenstein. A lattice computation returns $[\mathcal{O} : A_{15}(155)] = 5^{17} \cdot 31$, $e_5 = 2$, layers $(1, 12, 15)$ and type 5 (§9). The class number, by contrast, can break the almost Gorenstein equality ($A_{40}(451)$, §6.1).

Composite q' . For prime q' the condition $-1 \in \langle 2 \rangle$ says that t is even; for composite q' it says more. For $q' = 15$, $t = 4$ but $2^2 = 4 \neq -1$, so $\delta = 1$: for $A_7(105)$ at $p = 7$ the corollary gives

$W_1 = 3$, $e = 2$, $v_7([\mathcal{O} : A]) = v_7([A : \mathfrak{f}]) = 4$ and a Gorenstein ring, and a lattice computation returns $[\mathcal{O} : A_7(105)] = 7^4$, layers $(1, 3, 4)$ and type 1. For $q' = 63$, $t = 6$ and $\delta = 3$: the lattice of $A_{31}(315)$ at $p = 5$ has index 5^{20} , layers $(1, 15, 18)$ and type 5, as predicted. The hypothesis on h^- is needed: with $p \mid h^-$ the prediction fails in 3 of 7 controls, for instance $(q', p) = (77, 5)$, where $W_1 = 28$ and not 29. *Measured*: the layers W_0, W_1, W_2 agree with the corollary in all 1038 cases computed, over odd $5 \leq q' < 90$ other than 81, $p \leq 31$ and $N \leq 5$, with 13 composite values of q' at which t is even and $-1 \notin \langle 2 \rangle$; the type $2\delta - 1$, computed from the layers, agrees in the 26 cases computed, with $\delta \leq 4$ (§9).

Both corollaries end in the same two numbers, and so does the case $W_1 = d$. Once the first layer is more than half full, the index is settled.

Corollary 18 (the index). *Under the hypotheses of Theorem 5, suppose $W_1 > d/2$, or $W_1 > 0$ and $W_2 = d$. Then $e \leq 2$ and*

$$v_p([\mathcal{O} : A]) = 2d - 1 - W_1,$$

while $v_p([A : \mathfrak{f}]) = 1$ if $W_1 = d$ and $1 + W_1$ if $W_1 < d$. The hypothesis holds in the families of Theorem 12(a) as soon as $v_p(h^-(\mathbb{Q}(\zeta_{q'}))) < d/2$, and there $v_p([\mathcal{O} : A]) = d - 1 + k$ with k the number of elementary divisors of $M_{q'}$ divisible by p ; it holds always in Corollary 17, where $v_p([\mathcal{O} : A]) = d - 1 + \delta$, with $\delta = 0$ in case (a). Hence, if every prime $p \mid q$ either has q'_p dividing none of $h, h + 1, h + 2$ or satisfies the hypotheses of this corollary,

$$[\mathcal{O} : A] = \prod_p p^{d_p - 1 + (d_p - W_1(p))},$$

the product over the primes $p \mid q$ with q'_p dividing $h, h + 1$ or $h + 2$, and $d_p = \varphi(q'_p)/2$.

Proof. If $W_1 = d$ then $e = 1$ by Theorem 5(c) and Theorem 5(d) gives $v_p([\mathcal{O} : A]) = d - W_0 = d - 1$, which is $2d - 1 - W_1$. If $W_1 < d$ then $e = 2$: by Proposition 8(b) when $d/2 < W_1$, and by Proposition 6(b) when $W_1 > 0$ and $W_2 = d$. Theorem 5(d) then gives $(d - 1) + (d - W_1)$ and $1 + W_1$. In Theorem 12(a), $d - W_1 = k$ and each of those k elementary divisors contributes at least one factor p to $\Delta_{q'}$, so $k \leq v_p(\Delta_{q'}) = v_p(h^-)$ by Proposition 13 (at $r = q'/2$ through Lemma 14(ii) when $q' \equiv 2 \pmod{4}$), and $v_p(h^-) < d/2$ gives $W_1 > d/2$. In Corollary 17, $W_1 = d$ in case (a), and $d - W_1 = \delta = d/t$ with $t \geq 3$ in case (b), so $W_1 \geq 2d/3$. For the last formula, primes $p \nmid q$ do not divide the index by Theorem 4(a). A prime $p \mid q$ with q'_p dividing none of $h, h + 1, h + 2$ has $q'_p \notin \{1, 2, 3, 4\}$, since h is even, 4 divides h or $h + 2$, and 3 divides one of three consecutive integers; if $q'_p \neq 6$, Theorem 4(c) applies. If $q'_p = 6$, then $m \equiv 1 \pmod{3}$, the stabiliser in Theorem 4(b)(i) is trivial because $(\mathbb{Z}/6)^\times / \pm 1$ is, and counting as in the proof of Theorem 4(c) gives $S(1) = 2k + 1$ and $S(2) = -2k$ or $-2(k + 1)$ for $m = 6k + 1$ or $m = 6k + 4$; these are coprime, so (ii) holds and $p \nmid [\mathcal{O} : A]$. $\square$

Measured against the lattice: on the 104 pairs (order, prime) of the index sweep with p odd, $q' \notin \{1, 2, 3, 4, 6\}$ and $W_1 > d/2$, the formula gives the exponent of the lattice index in every case, and the explicit form does so in the 103 of them with $p \nmid N$ (or $p \nmid B$) and, in the centred family, $p \nmid h^-$; the three other pairs of the sweep with p odd and $q' \notin \{1, 2, 3, 4, 6\}$ all have $W_1 = 0$ and $p \mid N$, where Propositions 19 and 20 take over. Figure 8 places the measured profiles on the first-layer axis: of the 168 with $q' \notin \{1, 2, 3, 4, 6\}$, 126 have $W_1 = d$ and 18 have $d/2 < W_1 < d$, so the first clause decides them; the 17 with $0 < W_1 \leq d/2$ all have $p = 2$, and the second clause decides the 10 of them whose W_2 is within the computed range, always d (in the other 7, $E = 2$, and the lattice lengths are those of $e = 2$); the remaining 7 have $W_1 = 0$ (§9).

▷ *Theorem 12 lives on one line of one group. On the principal line of an exceptional group, is there an order at which a prime dividing a relative class number enters the conductor, and which Euler factor then plays the role of $\bar{\chi}(2) - 1$?*

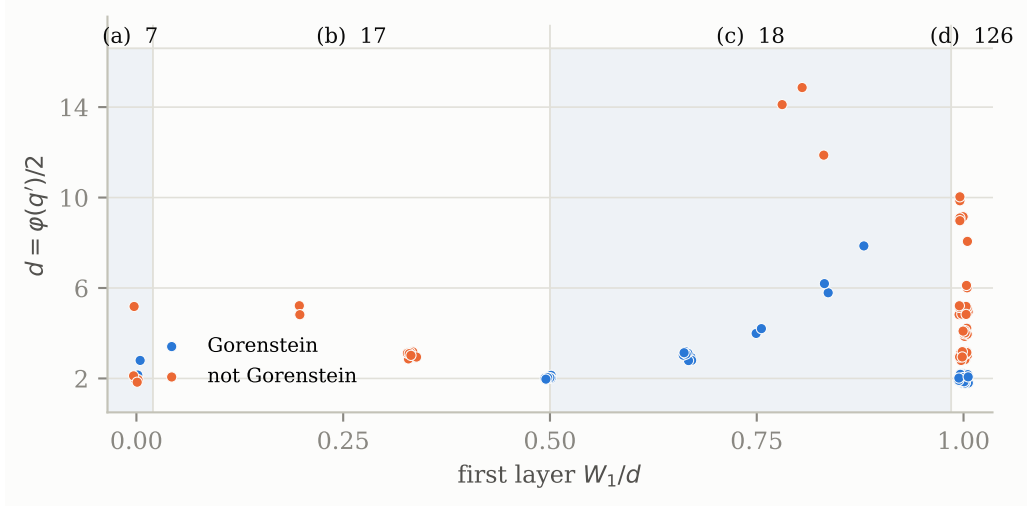


Figure 8: Where the first layer falls, on the 168 measured profiles with $q' \notin \{1, 2, 3, 4, 6\}$, and what decides the exponent in each band: (a) $W_1 = 0$, the zone of the ladder and the descent (Propositions 19 and 20); (b) $0 < W_1 \leq d/2$, where Corollary 18 needs its second clause $W_2 = d$; (c) $d/2 < W_1 < d$, where Proposition 8(b) gives $e = 2$; (d) $W_1 = d$, where $e = 1$. Colour marks whether the local ring is Gorenstein. The vertical axis is $d = \varphi(q')/2$, with a small jitter so that repeated pairs separate.

7 What lies beyond the first layer?

The first layer is now tied to class numbers, and it fixes the exponent when it is full or more than half full. What do the deeper layers look like?

Observed and certified. A finite window of layers does not decide the exponent in general: $A_7(45)$ at $p = 3$ has $(W_1, \dots, W_5) = (0, 1, 2, 1, 2)$ with $d = 2$, so a full layer can be followed by a deficient one; its lattice computation gives $\mathfrak{f}\mathcal{O}_3 = \mathfrak{r}^5$. Call *observed exponent* the first level from which all computed layers are full. In 1183 pairs (order, prime) whose layers were computed, most of them without a lattice computation ($p \leq 7$, $q' \leq 24$, $q \leq 260$, layers $i \leq 6$), all 215 with $1 \leq W_1 < d$ and at least four computed layers have observed exponent 2 (34 more have too few layers to say). In all 215, Proposition 6(b), applied to the computed $W_1 > 0$ and $W_2 = d$, certifies $e = 2$, also in the 10 with several primes above p . What makes that second layer full is the ring structure alone: *measured* on 238 pairs with p odd and $p \nmid N$, $\text{gr}_2(A) = \text{gr}_1(A) \cdot \text{gr}_1(A)$, so no new generator appears in degree 2, and that product is the whole of $\bar{R}$ in all of them. It is not full *because* the layer is nonzero, though: the square of a stable line is again a line, and the ladder produces exactly that, $\text{gr}_{P-p^j}(A)$ being a line by Proposition 19, which is why the exponent is large there. So the argument of Corollary 18 does not extend to the hypothesis $\text{gr}_1(A) \neq 0$: a nonzero first layer need not square to $\bar{R}$. For $A_4(40)$ at $p = 2$ one has $q' = 5$, $d = 2$ and $S(c) \equiv 1 \pmod{2}$ for $c \neq 0$, so $W_1 = 1$ and the square of that line is a line; its second layer is full all the same, $W = (1, 1, 2)$, through a new generator in degree 2. Whether the corollary itself survives that weakening we do not know; among the lattice profiles, the 10 with $0 < W_1 \leq d/2$ whose second layer was computed, all at $p = 2$, have $W_2 = d$. On the families of Theorem 12(a) its first condition, $W_1 > d/2$, says that fewer than $d/2$ elementary divisors of $M_{q'}$ are divisible by p ; since that number is at most $v_p(h^-)$, the bound $v_p(h^-) < d/2$ suffices, and to do without it this argument would need a bound on that number of elementary divisors, which Proposition 13 does not give. *Measured*: among 359 stable subspaces of $\bar{R}$ generated by random vectors, over $q' \leq 25$ and $p \leq 11$, every one with $\dim V > d/2$ squares to $\bar{R}$, while 7 of the 9 with $\dim V \leq d/2$ do not (§9). For $p \in \{2, 3, 5\}$, $q' \in \{5, 7, 9, 11, 13\}$ and layers $i \leq 11$,

the computed profile depends only on p, q' , the family and $v_p(N)$, and the observed exponent is often, but not always, $e_0 p^{v_p(N)}$, where e_0 is the observed exponent with $v_p(N) = 0$.

	1	2	3	4	5	6	7	8
$p=3, q'=5 \ (m=7, q=45)$	0/2	1/2	2/2	1/2	2/2			
$p=3, q'=7 \ (m=10, q=63)$	0/3	1/3	2/3	3/3	2/3			
$p=3, q'=11 \ (m=16, q=99)$	0/5	1/5	5/5	5/5	5/5			
$p=3, q'=13 \ (m=19, q=117)$	0/6	1/6	6/6	5/6	6/6			
$p=5, q'=7 \ (m=17, q=175)$	0/3	0/3	0/3	1/3	2/3	3/3	3/3	3/3
$p=5, q'=9 \ (m=22, q=225)$	0/3	0/3	0/3	1/3	3/3	3/3	2/3	3/3
$p=5, q'=11 \ (m=27, q=275)$	0/5	0/5	0/5	1/5	5/5	4/5	4/5	5/5
$p=5, q'=13 \ (m=32, q=325)$	0/6	0/6	0/6	1/6	6/6	6/6	6/6	6/6
$p=7, q'=3 \ (m=10, q=147)$	0/1	0/1	0/1	0/1	0/1	1/1	1/1	1/1
$p=7, q'=5 \ (m=17, q=245)$	0/2	0/2	0/2	0/2	0/2	1/2	2/2	2/2

Figure 9: Layers W_i , written as W_i/d , for ten orders with p odd and $v_p(N) = 1$. Every layer of positive degree below $p-1$ vanishes and layer $p-1$ (framed) has dimension 1, as Proposition 19 proves when $q' \geq 5$. Only layers $i < E = \varphi(p^v)$ were computed, which is why the rows with $p=3$ ($E=6$) stop at $i=5$. The rows show layers, not certified exponents (Proposition 6).

What the figure already certifies. Corollary 7 turns some rows of Figure 9 into exponents without computing further layers. For $A_{10}(63)$ at $p=3$ the layers are $(1, 0, 1, 2, 3, 2)$ with $d=3$: $W_2, W_3 > 0$ and $W_4 = d$, and $\langle 2, 3 \rangle$ contains every integer ≥ 2 , so every layer from 4 on except possibly 5 is full; $W_5 = 2$, so $e = 6$, $\mathfrak{f}\mathcal{O}_3 = \mathfrak{r}^6 = 3\mathcal{O}_3$, and by Proposition 10 $A_{10}(63) \otimes \mathbb{Z}_3$ is Gorenstein. In the same way the rows give $e = 5$ for $A_7(45)$, $e = 3$ for $A_{16}(99)$ and $e = 5$ for $A_{19}(117)$ at $p=3$, and $e = 8$ for $A_{22}(225)$ and $A_{27}(275)$ and $e = 5$ for $A_{32}(325)$ at $p=5$; the rows of $A_{17}(175)$ and $A_{17}(245)$ only bound e , and $A_{10}(147)$ has $q' = 3$, outside Theorem 5. Lattice computations confirm the five values that are within reach ($A_7(45)$, $A_{10}(63)$, $A_{16}(99)$, $A_{19}(117)$, $A_{22}(225)$).

The first zone of these profiles is not an accident of the examples: before degree $p^{v_p(N)}$ it is universal.

Proposition 19 (a universal ladder). *Let $A \neq \mathbb{Z}$, let p be odd, $q' \geq 5$ odd, $2m+1 = Nq'$ with $N = p^a M$, $a \geq 1$, $p \nmid M$, and $q = p^v q'$; put $P = p^a$. Then $v > a$, because $A \neq \mathbb{Z}$ forces $q > 2m+1$ by Proposition 2, that is $p^a M < p^v$; and $W_0 = 1$ and, for $1 \leq i < P$,*

$$W_i = \begin{cases} 1, & i \in \{P - p^j : 0 \leq j < a\}, \\ 0, & \text{otherwise.} \end{cases}$$

In particular the first nonzero layer of positive degree is $W_{\varphi(P)} = 1$, and the image of A_p in $\mathcal{O}_p/\mathfrak{r}^P$ is the ring

$$\mathbb{F}_p[\epsilon_0, \dots, \epsilon_{a-1}] / (\epsilon_i \epsilon_j : 0 \leq i, j < a),$$

with ϵ_j in degree $P - p^j$.

Proof. Put $Q_m(T) = \prod_{j=-m}^m (T - \xi^j) = (T-1) T^m \prod_{j=1}^m (T + T^{-1} - \eta_j)$; the passage between the coefficients of Q_m and the e_k is triangular over $\mathbb{Z}$, so they generate the same ring A .

Work in $\mathbb{Z}[\xi]$, unramified over $\mathcal{O}$ above p , with $\xi = \omega\rho$, ω of order q' , ρ of order p^v and uniformiser $\pi = \rho - 1$. Since $E = \varphi(p^v) \geq P$, layers below P only see the image of A in $\mathbb{Z}[\xi]/(p, \pi^P) = \mathbb{F}_p[\omega] \otimes \mathbb{F}_p[\rho]/(\rho^P - 1)$, where ξ has order $q'P$. The $2m+1 = Mq'P$ consecutive exponents cover every residue modulo $q'P$ exactly M times, so the coefficients of Q_m are invariant under $(\mathbb{Z}/q'P)^\times$, which acts on the two factors separately. The invariants of $\mathbb{F}_p[\omega]$ are $\mathbb{F}_p$; those of $\mathbb{F}_p[\rho]/(\rho^P - 1)$ are spanned by the sums over the subgroups $p^j\mathbb{Z}/P$, $\sum_{p^j|i} \rho^i = (\rho^P - 1)/(\rho^{p^j} - 1) = \pi^{P-p^j}$, $0 \leq j \leq a$, the last of which is 1. The products of the a nonconstant ones, $0 \leq i, j < a$, vanish modulo π^P because $p^i + p^j \leq 2P/p < P$, which gives the upper bound and the ring structure. For the lower bound take one block, $L = q'P$: by the q -binomial theorem the coefficients of $\prod_{j=0}^{L-1} (T - \xi^j)$ are, up to units, $\begin{bmatrix} L \\ k \end{bmatrix}_\xi = \prod_{i=1}^k (1 - \xi^{L-k+i})/(1 - \xi^i)$. Here $1 - \xi^i$ is a unit unless $q' \mid i$, and then $v_\pi(1 - \xi^i) = p^{v_p(i)}$ because $v_p(i) \leq a < v$. Pairing i with $L - i$, which have the same $v_\pi(1 - \xi^i)$ for $0 < i < L$ (if $q' \mid i$ then $v_p(i) < a$), leaves $v_\pi \begin{bmatrix} L \\ k \end{bmatrix}_\xi = P - p^{v_p(k)}$ if $q' \mid k$ and P otherwise. For $k = q'p^j$ this is $P - p^j$. Modulo π^P the full product is the M -th power of a shifted block, $(T^L - 1 + X)^M \equiv (T^L - 1)^M + M(T^L - 1)^{M-1}X$, since the coefficients of X have valuation at least $P - p^{a-1} \geq P/2$; its coefficient of $T^{q'p^j}$ is $\pm M$ times that of X , of valuation exactly $P - p^j$, and M is a unit. $\square$

For $a = 1$ this is the observation $W_{p-1} = 1$ of Figure 9; for $a = 2$ and $p = 3$ the nonzero layers of positive degree below 9 are W_6 and W_8 . The first nonzero term is itself a classical object: *measured* in 16 cases, any block of $q'P$ consecutive exponents, here the centred one, satisfies

$$\prod_{|j| \leq (q'P-1)/2} (T - \xi^j) \equiv T^{q'P} - 1 + \pi^{\varphi(P)} \text{li}_1(T^{q'p^{a-1}}) \pmod{\pi^{\varphi(P)+1}}, \quad \text{li}_1(X) = \sum_{r=1}^{p-1} \frac{X^r}{r},$$

the finite logarithm of Kontsevich, in Besser's notation [4].

At degree $p^{v_p(N)}$ itself the arithmetic of the first moment comes back, as the first layer of a smaller order of the same family.

Proposition 20 (descent to the reduced segment). *Under the hypotheses of Proposition 19, put $m_0 = (Mq' - 1)/2$, $q_0 = q/P$ and $A_0 = A_{m_0}(q_0)$. Then*

$$W_P(A_m(q)) = W_1(A_0) = \dim_{\mathbb{F}_p} \mathbb{F}_p[(\mathbb{Z}/q')^\times] \cdot S_0, \quad S_0(c) = M\hat{c},$$

both layers taken at p . Moreover, with $w = W_1(A_0)$, the ring $A_p/(A_p \cap \mathfrak{r}^{P+1})$ is $\mathbb{F}_p$ plus an ideal of square zero with a basis $\epsilon_0, \dots, \epsilon_{a-1}, \delta_1, \dots, \delta_w$, of filtration degrees $P - p^j$ and P .

Proof. Keep the notation of the proof of Proposition 19, with $n = q'$ and $k = \mathbb{F}_p[\omega] = \mathbb{F}_p[X]/(\overline{\Phi}_n(X))$, $\omega = X$: a product of copies of $\mathbb{F}_{p^{f'}}$, with f' the order of p modulo n , one for each prime of $\mathbb{Q}(\zeta_n)$ above p ; it need not be a field, and only its being étale is used below. Write ι for the involution $\omega \mapsto \omega^{-1}$ of k and $k^\pm$ for its ± 1 -eigenspaces, so that $k = k^+ \oplus k^-$ because $p \neq 2$. As $v \geq a + 1$, $E \geq 2P \geq P + 1$, so $p \in \pi^{P+1}$ and $\Lambda = \mathbb{Z}_p[\xi]/(\pi^{P+1}) = k[\pi]/(\pi^{P+1})$; the image B of A_p in Λ is the $\mathbb{F}_p$ -algebra generated by the coefficients of Q_m , and for $i \leq P$, $\text{gr}_i(A)$ is the image of $B \cap \pi^i \Lambda$ in $\pi^i \Lambda / \pi^{i+1} \Lambda = k$.

Blocks. Put $\ell = Mn = 2m_0 + 1$ and $b = (P - 1)/2$. Every integer in $[-m, m]$ is $c + \ell j$ for exactly one pair $|c| \leq m_0$, $|j| \leq b$. With $\tau = \xi^\ell = \rho^\ell$, a primitive p^v -th root of unity, write $\prod_{|j| \leq b} (Z - \tau^j) = Z^P - 1 + \sum_{k=1}^{P-1} b_k Z^{P-k}$. Then

$$Q_m(T; \xi) = \prod_{|c| \leq m_0} (T^P - \xi^{cP} + E_c(T)), \quad E_c(T) = \sum_{k=1}^{P-1} b_k \xi^{ck} T^{P-k}, \quad (7)$$

and the valuation argument of Proposition 19, applied to $\left[\frac{P}{k}\right]_\tau$, gives $v_\pi(b_k) = P - p^{v_p(k)} \geq s := P - P/p$, with $2s > P$. So, modulo π^{P+1} , only the terms with at most one factor E_c survive. With $\zeta = \omega^P$, $\xi_0 = \xi^P$ and $F_c(Y) = \prod_{c' \neq c} (Y - \xi_{c'})$, modulo π^{P+1}

$$[T^{Pj}] Q_m(T; \xi) \equiv [Y^j] Q_{m_0}(Y; \xi_0), \quad [T^{Pj+P-k}] Q_m(T; \xi) \equiv b_k \sum_{|c| \leq m_0} \xi^{ck} [Y^j] F_c(Y).$$

Partial fractions. Let $F(Y) = \prod_{|c| \leq m_0} (Y - \zeta^c) = (Y^n - 1)^M$ and $G_h = \sum_{c \in \mathbb{Z}/n} S_0(c) \zeta^{ch}$. Grouping c by residues, $\sum_{|c| \leq m_0} \zeta^{ch} F(Y)/(Y - \zeta^c) = Mn(Y^n - 1)^{M-1} Y^{i_h}$ with $i_h \equiv h - 1$, an integer polynomial, and $D_h(Y) = \sum_{|c| \leq m_0} c \zeta^{ch} F(Y)/(Y - \zeta^c) = (Y^n - 1)^{M-1} \sum_{i < n} G_{h-1-i} Y^i$. Let $V \subseteq k$ be the span of the residues of the G_h ; the coefficients of every D_h lie in V , and those of D_1 in degrees $n(M-1) + i$ are the G_{-i} , so they span V . As S_0 is odd, G_h is anti-invariant under $\omega \mapsto \omega^{-1}$; so $V \subseteq k^-$, while $\mathbb{F}_p \subseteq k^+$ and $k^+ \cap k^- = 0$.

The order A_0 . It satisfies the hypotheses of §5 at p : $q_0 = p^{v-a}n$ with $E_0 \geq 2$, $n \mid 2m_0 + 1$, and $2m_0 < q_0$ because $PMn \leq p^v n$ and $p \nmid M$. With $\pi_0 = \rho^P - 1$, $Q_{m_0}(Y; \xi_0) \equiv F(Y) - \pi_0 D_1(Y) \pmod{\pi_0^2}$, so, exactly as for A_p above, $\text{gr}_1(A_0) = V$ and $W_1(A_0) = \dim V$, which Theorem 5(b) computes as $\dim \mathbb{F}_p[(\mathbb{Z}/n)^\times] \cdot S_0$.

Generators. Since $\pi_0 \equiv \pi^P$ modulo π^{P+1} , the coefficients in degrees Pj are $[Y^j]F - \pi^P[Y^j]D_1$, with $[Y^j]F \in \mathbb{Z}$. For $k = p^t u$, $p \nmid u$, choose h with $Ph \equiv k \pmod{n}$; modulo π^{P+t} one has $F_c \equiv F/(Y - \zeta^c)$ and $\xi^{ck} \equiv \zeta^{ch}(1 + cu\pi^{p^t})$. Writing $b_k = \pi^{P-p^t}\beta$ with $\beta \equiv \beta_k \in \mathbb{Z} \pmod{\pi}$, the coefficients in degrees $Pj + P - k$ are $N_{k,j} b_k + \beta_k u \pi^P [Y^j] D_h$ with $N_{k,j} \in \mathbb{Z}$.

Lower bound. The elements $\pi^P [Y^j] D_1 = [Y^j]F - [T^{Pj}]Q_m$ lie in $B \cap \pi^P \Lambda$ with residues spanning V , so $V \subseteq \text{gr}_P(A)$.

Upper bound. Every generator lies in $C = \Lambda_\rho + \pi^P V$, where Λ_ρ is the image of $\mathbb{Z}_p[\rho]$; C is a ring, so $B \subseteq C$, and an element of C in $\pi^P \Lambda$ has its Λ_ρ -part in $\mathbb{F}_p \pi^P$. Hence $\text{gr}_P(A) \subseteq \mathbb{F}_p + V$, which is one dimension too many.

The scalar direction. It is complex conjugation that removes it, and this is where the order being real is used. Conjugation fixes A_p pointwise and sends π to $-\pi\rho^{-1}$; as P is odd, the residue y of an element of $A_p \cap \mathfrak{r}^P$ satisfies $y = -\bar{y}$, that is $\text{gr}_P(A) \subseteq k^-$, while $\mathbb{F}_p \subseteq k^+$ and $k^+ \cap k^- = 0$ because $p \neq 2$. So $\text{gr}_P(A) \subseteq (\mathbb{F}_p + V) \cap k^- = V$, and with the lower bound $\text{gr}_P(A) = V$.

Ring. Each generator is an integer plus an element of $\pi^s \Lambda$, and $2s > P$, so products of these elements vanish, $B = \mathbb{F}_p + \mathfrak{m}$ with $\mathfrak{m}^2 = 0$, and $\dim \mathfrak{m} = \sum_{1 \leq i \leq P} W_i = a + w$ by Proposition 19, so $\dim_{\mathbb{F}_p} B = 1 + a + w$. $\square$

Reality is used, and needed, in the last bound: for the non-centred product $\prod_{0 \leq x \leq 2m} (T - \xi^x)$, which is not real, π^P does lie in the algebra of its coefficients modulo π^{P+1} (*measured* in four cases). The proposition holds in the 32 + 953 cases computed, and its equality $W_P = W_1(A_0)$ in 118 further ones. The 953 run over $p \leq 11$, $p^{v_p(N)} \leq 121$, $M \leq 13$ and odd $q' \leq 45$ other than 37, 41, 43, 338 of them with composite q' ; 51 cases of that grid above a cost bound were skipped, and a second value of v was tried only on the cheaper cases. For $q' = 7$ it gives the rank 2 of the running example, which is the shortfall of W_p in Figure 9.

Question 21. Proposition 20 recovers, at degree $p^{v_p(N)}$, the first layer of the reduced order. Does the profile beyond it keep descending, layer $P + i$ of $A_m(q)$ being governed by layer $1 + i$ of $A_{m_0}(q_0)$ corrected by the ladder, and does it continue as a ladder of generalised Bernoulli numbers $B_{i,\chi}$ modulo $\mathfrak{p}$, so that the p -part of $[\mathcal{O} : A]$, which by Theorem 5(d) is the sum of the defects $\sum_{i < e} (d - W_i)$, has a closed form in terms of p -adic L -functions?

Measured beyond the descent, on the 38 pairs with $p \in \{3, 5, 7\}$, $q' \leq 27$, $M \leq 4$ and $v_p(N) = 1$: the profile does not simply descend, since layer $P + 1$ is not always the second layer of A_0 , which is full whenever $W_1(A_0) > d/2$ (Proposition 8 with $i = j = 1$) and was full in all 238

pairs measured above. What appears to govern it is the *second* moment. Writing $\hat{c}$ for the signed residue, the segment of a centred family has $\sum_{|j| \leq m_0, j \equiv c} j^2 = M\hat{c}^2 + M(M^2 - 1)q'^2/12$, an even function, and W_{P+1} is the dimension of the cyclic $\mathbb{F}_p[(\mathbb{Z}/q')^\times]$ -module generated by $c \mapsto \hat{c}^2$ in 33 of the 38 pairs, and one more than it in the remaining 5; while $W_{P+2} = W_P$ in 26 of the 38, all eight with $p = 3$ among them. So, in these pairs, the first moment decides the first layer, and the second moment accounts for the layer above the descent in 33 of the 38; that is an observed pattern, not a theorem, and it points to the B_2 end of the ladder the question asks about. What the 5 exceptions add, we do not know (§9).

▷ Along the tower $q = p^v q'$ with $m = p^{v-1} q' - 1$, does $v_p([\mathcal{O} : A])$ grow like dp^{v-1} plus a term linear in v , and is the coefficient of v an Iwasawa invariant of $\mathbb{Q}(\zeta_{q'p^\infty})$?

8 Conclusions and limits

What has been proved, what rests on the work of others, what was only measured, and where does the method stop?

Proved here. The support of the conductor (Theorem 4); its local structure, the lengths as sums of layer defects, the growth and complementarity of layers, and the Gorenstein criterion in layers (Theorem 5, Propositions 6–10); and the link between the first layer and h^- for $m \equiv 0, -1 \pmod{q'}$ with p odd, $p \nmid N$ and $q' \notin \{1, 2, 3, 4, 6\}$, including $q' \equiv 2 \pmod{4}$, and for the half-period families $m \equiv r, r-1 \pmod{q'}$ when $q' \geq 8$ is even, $r = q'/2$ and $p \nmid \lceil m/r \rceil$ (Lemma 14), with its consequence that a simple divisor of h^- gives the square of the radical at p and a Gorenstein local ring (Theorem 12, Corollary 16); the factor at 2 without semisimplicity, for prime and composite odd q' , with the type of the resulting almost Gorenstein rings (Corollary 17, Proposition 11); certification by semigroups (Corollary 7); the index at a prime whose first layer is more than half full (Corollary 18); the universal ladder below $p^{v_p(N)}$ and the descent at $p^{v_p(N)}$ (Propositions 19 and 20). The symmetric form of the Gorenstein criterion, and the complementary inequality in degree $e-1$, are the diagonal case of the criterion of Campillo, Delgado and Kiyek [9], in the line of Kunz’s symmetric value semigroups [20, 11]; the growth of layers (Proposition 6) and the complementary inequality in other degrees we have not found in that literature.

Resting on others. Amiot’s lemma on scales; Sinnott’s index formula in the form of Bernard and Kučera, of which Proposition 13 is a restatement in matrix form; and the characterisation of Gorenstein rings by lengths (Bass, Serre, Grothendieck, Herzog–Kunz, Huneke–Swanson) with its descent; the formula $\text{type} + 1 = \dim(\mathfrak{m} : \mathfrak{m})/\mathfrak{m}$ of Marseglia, of which Proposition 11 is a direct consequence, and the almost Gorenstein rings of Barucci and Fröberg. The rank of M_n over $\mathbb{Q}$ is Kuribayashi’s, and the Euler factor $\bar{\chi}(2) - 1$ of the shifted sawtooth is in Kanemitsu and Kuzumaki and in Kučera. We have not found an earlier statement of Proposition 13 as a formula for the maximal minors of M_n , nor of Lemma 14(ii) in this matrix form (it is the distribution relation of B_1), nor of Corollary 17(b) or of its part (a) outside Kučera’s Example 2, and we claim priority for none of them.

What the identification does not say. For odd $p \nmid 2q'$ it identifies the p -part of the cokernel of the moment matrix with that of R^-/S^- ; it does not identify $\mathcal{O}_p/A_p$, as a Galois module, with a piece of the class group. It decides the first layer, and with it the exponent when $W_1 = d$ or $W_1 > d/2$; below that, the exponent depends on deeper layers that no theorem here computes when $p \nmid N$. When $v_p(h^-) \geq 2$, $v_p(\Delta_{q'})$ is known but neither the lengths nor how the p -parts of the elementary divisors are distributed, nor whether that distribution reaches those layers.

Not covered. $q' \leq 2$, i.e. $q = p^v$ or $2p^v$, where $\omega = \pm 1$ and the first-order term vanishes; $q' \in \{3, 4, 6\}$, where Theorem 4(b) still applies but the statements of §5 need not; $p = 2$ in Theorem 12, where for $m \equiv 0, -1 \pmod{q'}$ with N odd one has $S(c) \equiv 1$ for $c \neq 0$, so $W_1 = 1$ whatever h^- is; $p \mid N$ (on the half-period families, $p \mid \lceil m/r \rceil$) in Theorem 12(a), and beyond

degree $p^{v_p(N)}$ in the family $q' \mid h+1$; the family $q' \mid h+1$ when $p \mid h^-(\mathbb{Q}(\zeta_{q'}))$ and either q' is composite or $p \mid q' - 1$; the type when $e \geq 3$; and the index at a prime with $W_1 \leq d/2$ and $W_2 < d$, beyond Corollary 18, where, for p odd, $p \mid N$ and $2m+1 = Nq'$, only the ladder and the descent speak.

9 How the numbers were computed

Orders. For each (m, q) the order A is built in Sage as a lattice in $\mathcal{O} = \mathbb{Z}[\alpha]$, $\alpha = \xi + \xi^{-1}$, by closure: starting from $\mathbb{Z} \cdot 1$, the lattice is replaced by the Hermite normal form of itself plus its products with $e_1, \dots, e_m$ until it stops changing. The fixed point contains 1 and is stable under multiplication by the e_i , so it contains $\mathbb{Z}[e]$; every element of it is obtained from 1 by such products and sums, so it is contained in $\mathbb{Z}[e]$. On 22 small cases this lattice coincides with Sage's own `order` constructor.

Conductor and index. $[\mathcal{O} : A]$ is the determinant of the basis matrix in powers of α . The conductor is the lattice $\{x \in \mathcal{O} : x\alpha^k \in A, 0 \leq k < D\}$, $D = [K : \mathbb{Q}]$, computed as the integer kernel of the corresponding linear conditions, and it is factored after maximising only at the primes of its norm.

Layers. W_i is computed without the conductor. Let Ψ_n be the minimal polynomial of $\zeta_n + \zeta_n^{-1}$, of degree $\varphi(n)/2$. Since $\mathcal{O} = \mathbb{Z}[\alpha]$ and $\Psi_q \equiv \Psi_{q'}^E \pmod{p}$, for $i < E$

$$\mathcal{O}/(p\mathcal{O} + \mathfrak{r}^{i+1}) \cong \mathbb{F}_p[X]/(\overline{\Psi}_{q'}(X)^{i+1}), \quad X \mapsto \alpha,$$

and W_i is the difference of the $\mathbb{F}_p$ -dimensions of the images of A at two consecutive truncations, obtained by linear algebra over $\mathbb{F}_p$, for $q' \geq 3$. (The scripts compute these images inside the ring $\mathbb{Z}[\xi]/(p, \Phi_{q'}(\xi)^{i+1}) \cong \mathbb{F}_p[y]/(\Phi_{q'}(y)^{i+1})$, which contains $\mathcal{O}/(p\mathcal{O} + \mathfrak{r}^{i+1})$ because $\mathbb{Z}[\xi]$ is unramified over $\mathcal{O}$ above p ; the dimensions of the images of A are the same.) Layers $i \geq E$ require p -adic precision and were not computed this way. On all 184 pairs (order, prime of the conductor) with a lattice computation and $e \leq E$, the formula $v_p([A : \mathfrak{f}]) = \sum_{i < e} W_i$ matches the lattice; one further pair has $e > E$, with $E = 1$. All 185 pairs have $q' \geq 3$, the range where the isomorphism above holds; 17 of them ($q' \in \{3, 4\}$ or $E = 1$, the pair with $e > E$ included) lie outside the hypotheses of §5. On the 168 pairs inside them whose layers below e were computed, the lemmas of §5 hold without exception, and the symmetry of Proposition 10 holds exactly when $v_p([\mathcal{O} : A]) = v_p([A : \mathfrak{f}])$.

Synthetic cases are pairs (order, prime) whose layers were computed, mostly without a lattice; their exponents are observed, and certified through Proposition 6(b) or, where the layers allow it, Corollary 7. The ranks of Figure 6 and Remark 15 are ranks of integer matrices reduced mod p . The relative class numbers of the older scripts are evaluated in floating point from the analytic class number formula and rounded. They are certified independently: for the 22 primes of Figure 6 by a product of norms from cyclotomic fields in rational arithmetic, and for all 66 values $n < 90$, $n \not\equiv 2 \pmod{4}$, by the same formula in exact cyclotomic arithmetic in Sage, which Figure 7 uses. PARI's class group computation agrees with proof for $\varphi(n) \leq 20$, and assuming GRH for $20 < \varphi(n) \leq 32$. The elementary divisors of Proposition 13 are computed over $\mathbb{Z}$ and never used to compute h^- .

Which cases, and where. The lattice computations are the pairs (m, q) listed in the header of each sweep file; only pairs with a result are used, and nothing is interpolated. All files sit in the flat directory `conductor/` of the archive.

numbers	script	output
orders, conductors, Figure 3	orden_generado.sage, factorizar_conductor.sage, h2_residuo.sage; gathered by datos_indice.py	celdas_m*_OUT.txt, h2_OUT.txt, r2_OUT.txt, datos_indice.json
closure = Sage order (22) layers; 184 pairs	orden_generado_control.sage perfil_W.py, ley_conductor.sage	orden_generado_control_OUT.txt perfil_W_OUT.txt, ley_conductor_ancho_OUT.txt, ley_conductor_repro_OUT.txt
1183 pairs, 215 observed 215 certified; lemmas of §5 on 168 lattices	catalogo_e.py capas_locales.py	catalogo_e_OUT.txt capas_locales_OUT.txt
profiles against $v_p(N)$ Figure 6, Theorem 12	catalogo_N.py bernoulli_W1.py, bernoulli_2m1.py	catalogo_N_OUT.txt bernoulli_W1_OUT.txt, bernoulli_2m1_OUT.txt
exact h^- (22 primes) Remark 15 (984)	h_menos_exacto.py rango_compuesto.py	h_menos_exacto_OUT.txt rango_compuesto_OUT.txt
$\Psi_q \equiv \Psi_q^E$; orbit ranks Proposition 13: 1505 pairs; exact h^- (66 values)	psi_y_orbitas.py stickelberger_smith.sage	psi_y_orbitas_OUT.txt stickelberger_smith_OUT.txt
h^- against PARI $A_{22}(69)$	h_menos_pari.sage, h_menos_pari_grh.sage orden_A22_69.sage	h_menos_pari_OUT.txt, h_menos_pari_grh_OUT.txt orden_A22_69_OUT.txt
semigroup certificates; lat- tices $A_{19}(117)$, $A_{22}(225)$, $A_{15}(155)$	item1.py, lat.sage	item1_OUT.txt, lat*_OUT.txt
ladder and W_P (118); fi- nite logarithm (16)	item2.sage, item2_log.py	item2_OUT.txt, item2_log_OUT.txt
factor at 2, prime q' (438)	item3.sage, item3_ctrl.sage	item3_OUT.txt, item3_ctrl_OUT.txt
type when $e = 2$ (403)	capas.sage, item4.sage, item4b.sage	item4_OUT.txt, item4b_OUT.txt
descent $(28 + 4)$; non- centred control (4)	check_descenso.py, control.py	out_batch.txt, out_hminus.txt, control_OUT.txt
ladder and descent (953) Lemma 14(ii): identities, lattices (22), layers (304)	inst.sage, descent.sage duplicacion.py, l2r.sage	descent_OUT_all.txt duplicacion_OUT.txt, l2r_OUT.txt

numbers	script	output
Lemma 14(i) (34); half period with $4 \mid q'$: layers (156 distinct; the output has 158 lines), controls (12); the identity ($n \leq 100$), count times midpoint (98), rank for general ℓ (52) with 7 controls at $p = 2$	medio.sage, cruce.sage	medio_OUT.txt, cruce_OUT.txt
Corollary 18 against the lattice indices (104; 103 explicit)	indice.sage	indice_OUT.txt
§7: second layer as a square (238); beyond the descent (38; $W_{P+2} = W_P$ in 26); stable subspaces and their squares (359)	capa2.sage, capa2_recuento.py, cuadrado.sage	capa2_OUT.txt, capa2_recuento_OUT.txt, cuadrado_OUT.txt
Corollary 17 for odd q' (1038), controls, type; $A_7(105)$, $A_{31}(315)$	dos.sage, dos_ctrl.sage, dos_tipo4.sage, latchk.sage	dos_OUT.txt, dos_ctrl_OUT.txt, dos_tipo4_OUT.txt, lat_A7_105_OUT.txt, lat_A31_315_OUT.txt
layer instruments against each other	xval_inst.sage, xval_inst_dimsB.py	xval_inst_OUT.txt, xval_inst_dimsB_OUT.txt
Kostant's element is $g_{1/(2m+2)}$; characters in $\{0, \pm 1\}$ there ($m \leq 5$)	kostant_check.py	kostant_check_OUT.txt
the 21 singular unit matrices; the bands of the 168 profiles; (5) on all 66	cifras_sin_salida.py	cifras_sin_salida_OUT.txt
all figures	figuras.py	fig_*.pdf

Declaration of competing interest

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Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used Claude (Anthropic) in order to draft prose, to write and run the verification scripts of §9, and to search the literature. After using this tool the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

Attribution

statement	source
a scale in $\mathbb{Z}/c$ with a coprime generator other than $\pm$ its own is full, almost full or trivial	Amiot [1], Thm. 1
$A = \mathbb{Z}$ (Proposition 2): $q \in \{2m + 1, 2m + 2\}$ or $(q, m) = (6, 1)$, the part $2m < q$ of the integer points orders $\mathbb{Z}[\chi(g)]$ for finite groups; primes of the index divide $ G $	[24], Thm. 5.1; rational elements, Kostant [18, Thm. 3.1] Bächle–Sambale [6], Thm. A, Prop. 6
conductor lengths; Gorenstein $\iff \ell(\mathcal{O}/A) = \ell(A/\mathfrak{f})$	Bass [7, Cor. (6.5)], with the converse after Serre and Grothendieck; Herzog–Kunz [16]; Huneke–Swanson [17], §12.2
symmetric form of the Gorenstein criterion (Proposition 10); Proposition 8 for $i + j = e - 1$	Campillo–Delgado–Kiyek [9], (3.6)–(3.7)
descent of the Gorenstein property; canonical modules	Stacks Project [26, Tag 0BJL]; Bruns–Herzog [8, Thm. 3.3.6, Prop. 3.3.18]
Maillet’s determinant, $D_p = \pm p^{(p-3)/2} h^-$	Carlitz–Olson [10], (1.7)
Stickelberger ideal and h^- ; its generators and indices (Proposition 13)	Sinnott [25, 27]; Bernard–Kučera [3]; Washington [29]
square determinants for composite conductors; cotangent index	Tateyama, Wang, Kučera [21]; Girstmair [14]
rank of the rectangular matrix M_n over $\mathbb{Q}$	Kuribayashi [22]; Breuer [5]
Euler factor $\bar{\chi}(2) - 1$ of the shifted sawtooth; Corollary 17(a) when no prime divisor of q' splits in $\mathbb{Q}(\zeta_{q'})/\mathbb{Q}(\zeta_{q'})^+$	Kanemitsu–Kuzumaki [19]; Kučera [21], (7), Ex. 2
type $+1 = \dim(\mathfrak{m} : \mathfrak{m})/\mathfrak{m}$; almost Gorenstein rings	Marseglia [23], Prop. 3.5, §7.2; Barucci–Fröberg [2]
finite logarithm	Kontsevich; Besser [4]
half sums of odd characters	Szmidt–Urbanowicz–Zagier [28], (6)
Lemma 3, Theorems 4, 5, 12, Propositions 6–9 (Proposition 8 beyond $i + j = e - 1$), Proposition 11 (from Marseglia’s formula), Propositions 19, 20, Lemma 14, Corollaries 7, 16, 17, 18	this note

Data availability

No external data were used. The computational data — every number labelled *measured* or *observed*, and every figure — are generated by the scripts described in §9, which accompany this note together with their archived output.

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